MANY TREATMENTS, SOME DISCRETE INSTRUMENTS: IDENTIFICATION AND INFERENCE FOR CAUSAL EFFECTS IN TREATMENT SELECTION MODELS

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PRELIMINARY AND INCOMPLETE

This paper presents new specification testing and partial identification results for instrumental variable (IV) models with multivalued treatment and randomly assigned discrete instruments. I develop a framework for deriving sharp bounds on a broad class of causal parameters under many different kinds of assumptions about the treatment selection mechanism. Other than random assignment of the instrument, I do not require any restrictions on treatment selection, but allow the analyst to flexibly model selection into treatment. I show that such models have identifying power *only* through restrictions on the joint distribution of counterfactual treatment choices ("response-type"). Using a novel geometric characterization of IV mixture models, I provide a characterization of the empirical content and testable implications of such models, and characterizations of the identified set for counterfactual choices, marginal and joint outcome distributions, outcome means, and treatment effects under such models. I propose methods for (i) (sharply) testing treatment selection model specifications and (ii) performing inference on mean causal effects and counterfactual outcomes under such models. Finally, for cases where analytical approaches are infeasible, I propose a computationally feasible algorithm for implementing the proposed method without relying on (semi)parametric assumptions.

1. INTRODUCTION

I study nonparametric instrumental variable (IV) models with multiple treatments, discrete instruments and heterogeneous agents. It is well-known that such models can (potentially after conditioning on baseline covariates) be represented as a generalized Roy model:

$$\text{Treatment Selection Equation: } D = g_D(Z, \boldsymbol{V}), \tag{1}$$

$$\text{Outcome Equation: } Y = g_Y(D, \boldsymbol{V}, \varepsilon_Y), \tag{2}$$

where Y is the observed (potentially continuous) outcome of interest, D represents the observed (discrete-valued) treatment status, Z represents the observed (discrete-valued) assigned instrument value, and $\boldsymbol{V}$ and ε_Y are (potentially infinite dimensional) unobservables representing unobserved heterogeneity; $Z, \boldsymbol{V}$ and ε_Y are mutually independent exogenous variables, which implies random assignment of the instrument Z. I require that D and Z are finitely-supported, but the support of Y is only required to be a compact convex subset of $\mathbb{R}^n$ (or finite). I also assume that the (generally unknown) functions g_D and g_Y satisfy some mild measurability and integrability conditions described below. Beyond these basic requirements, the analyst may be willing to make additional assumptions: I focus on models that impose

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restrictions on the treatment selection equation (1) but remain agnostic about the outcome equation (2)[1]. For such models, this paper provides

(i) a sharp characterization of their empirical content and testable implications
(ii) a new characterization of the identified set for counterfactual choices, outcome means, and treatment effects

For a broad class of such models, I also propose methods for inference based on the above identification results; in particular, this paper provides (sharp) specification tests for treatment selection models and inclusion tests for mean counterfactuals that can be inverted to construct confidence regions.

This paper's approach relies on partitioning the support of the latent variable $\boldsymbol{V}$ into *response-types*, based on how received treatment varies with assigned instrument value. The notion of response-types used in this paper is the same as Balke and Pearl (1994), Heckman and Pinto (2018), Gu and Russell (2022), and many other papers. I show that assumptions on g_D and/or $\boldsymbol{V}$ have identifying power *only* through restrictions on the distribution of response-types, although such restrictions do not have to be limited to which response-types are eliminated by the model. In an important paper, Heckman and Pinto (2018) showed that the problem of identifying response-type conditional counterfactuals can be represented as a finite mixture model where the mixture components are the conditional distributions generated by response-types. This paper attacks this mixture problem directly and develops new results on the partial identification of components of overlapping mixtures, which may be of independent interest.

This paper contributes to a large literature on identification in IV models, I only provide a very brief overview here. Much of the literature on IV models with multiple treatments and/or instruments have focused on point identification under relatively restrictive assumptions on treatment selection: Angrist and Imbens (1995) assume "monotonicity" of the instrument and extend Imbens and Angrist (1994)'s results to a setting with multiple levels of (ordered) treatments, Heckman, Urzua, and Vytlacil (2008) considers multiple treatments within the Marginal Treatment Effect (MTE) framework of Heckman and Vytlacil (1999, 2001, 2007a,b) but rely on continuous monotonic instruments, Heckman and Pinto (2018) consider point identification in the same setting as this paper and propose the "Unordered Monotonicity" condition which allows several mean counterfactuals to be identified, Mountjoy (2022) provides point identification for some parameters of interest using continuous instruments and the less restrictive "Unordered Partial Monotonicity" assumption, which builds on the "Partial Monotonicity" assumption introduced by Mogstad, Torgovitsky, and Walters (2023). The partial identification framework of this paper can sharply incorporate any of these assumptions on treatment selection.

Lee and Salanié (2018) consider a broad class of treatment selection models based on combinations of threshold crossing rules and develop point identification results that rely on continuous instruments. The identification results of the current paper apply to these models, but require knowledge of how such assumptions restrict the distribution of response-types, which may be generally non-trivial to derive. Even without this information, however, the framework can be used to obtain valid (potentially non-sharp) bounds.

More closely related to this paper is the growing literature on partial identification in IV models, as considered by Manski (1990); Manski and Pepper (2000, 2009), Shaikh and Vytlacil (2011), Mourifié (2015), Huber, Laffers, and Mellace (2017), Torgovitsky (2019b), among others. Within the MTE framework, Mogstad, Santos, and Torgovitsky (2018) proposed a method for partial identification in IV models with a mean independent monotonic

[1]Some types of assumptions on outcomes (e.g., mean dominance) can be incorporated, however.

instrument, and their approach was recently extended to more general selection models (partial monotonicity) by Mogstad et al. (2023). Gu and Russell (2023) propose a computational approach that can be used to obtain the identified region for counterfactuals in IV models under mean independence. Marx (2023) proposes an analytic approach to identification under random assignment of the instrument in a binary treatment setting with treatment selection governed by a threshold crossing rule, and Han and Yang (2023) proposes a computational approach for the same problem and setting. In a complementary recent paper, Kamat, Norris, and Pecenco (2023) proposed a computational approach for partial identification in the models considered by Lee and Salanié (2018) under random assignment of the instrument; their approach requires a semiparametric restriction on unobserved heterogeneity and they do not show (or claim) sharpness. The approach of this paper is different from all of these papers. Unlike many of these papers, this paper cannot accommodate continuous instruments or instruments that are only mean-independent, and generally cannot incorporate assumptions on the outcome equation. On the other hand, this paper provides sharp bounds under the stronger assumption of random assignment (rather than just mean independence) of the instrument, for a broader class of treatment selection models, and can accommodate multiple treatments and instruments without relying on any parametric or semiparametric assumptions, using a geometric framework that is less "black box" than purely computational approaches.

This paper is also related to the literature on specification testing in IV models. This paper extends the results of Kitagawa (2015) and Mourifié and Wan (2017) on sharp testable implications of LATE assumptions to treatment selection models beyond monotonicity and to the general multivalued treatment and instrument setting. As special cases, this paper also provides a sharp test for random assignment, for which a test was proposed by Kédagni and Mourifié (2020), and also for Unordered Montonicity, for which a test was proposed by Sun (2023). Neither of these papers show sharpness, however, unlike this paper[2]. The framework of this paper can also be used to perform sensitivity analysis in the spirit of Noack (2021), but I do not pursue this further in this draft. This paper is also related to the literature on the identification of joint distribution of potential outcomes, as investigated by Heckman and Smith (1995, 1998), Heckman, Smith, and Clements (1997), and more recently by Jun and Lee (2023) and Kaji (2023), but the approach of this paper is in a relatively much more general setting. Finally, this paper is related to the literature on nonparametric identification of mixture components, as in Horowitz and Manski (1995, 1998, 2000); Lee (2009) and Henry, Kitamura, and Salanié (2014).

Organization of the paper. The remainder of this paper is organized as follows. Section 2 formalizes our analytical framework and setting, including the notion of "treatment selection model", and develops some high level results about the source of identifying power in treatment selection models. Section 3 introduces our geometric characterization and uses it to obtain novel characterizations of the identified set for response-type probabilities and counterfactual means, and proposes a sharp specification test for treatment selection models. Section 4 outlines an inference strategy for implementing the proposed specification tests and develops an inclusion test for counterfactual means and treatment effects. Section 5 discusses potential computational challenges that may complicate practical implementation, and presents a generally applicable algorithm for computation of the objects necessary to employ the proposed methodology. Section 6 concludes. All but the shortest proofs are presented in the appendix.

[2]The results in this paper should allow us to resolve the the conjectured sharpness of these papers, but I have not yet pursued this.

Notation. For any set K, $|K|$ denotes its cardinality. For any number $n \in \mathbb{N}$, $[n] := \{1, \ldots, n\}$. For any event A, denote by $\mathbb{1}\{A\}$ the indicator function takes value 1 if A is true, and 0 otherwise. For any random variable X, $\text{Law}(X)$ denotes its distribution, and $\text{supp}(X)$ denotes its support. For any distribution F of some multi-dimensional variable $X = (X_1, \ldots, X_K)$: (i) $\text{supp}(F) := \text{supp}_F(X)$ denotes the support of X under $X \sim F$, (ii) F_{X_j} denotes the marginal distribution of X_j under $X \sim F$, (iii) $F_{X_j \mid X_k = x_k}$ denotes the conditional distribution of X_j, given $X_k = x_k$, under $X \sim F$.

2. IDENTIFICATION THEORY FOR TREATMENT SELECTION MODELS

2.1. **Basic Framework and Setting.** All analysis in this section and the next will treat the joint distribution of the observed data, (Y, D, Z), as known, and assume that it satisfies the following conditions. The support of D, denoted $\mathcal{D}$, and the support of Z, denoted $\mathcal{Z}$, are finite. The support of Y, denoted $\mathcal{Y}$, is either a finite set or a compact convex subset of $\mathbb{R}^n$, and equipped with the Borel σ-algebra $\mathscr{B}_{\mathcal{Y}}$ and a probability measure μ; Y has a density with respect to μ. For all $d \in \mathcal{D}$ and $z \in \mathcal{Z}$, $E[Y \mid D = d, Z = z]$ exists and $|E[Y \mid D = d, Z = z]| < \infty$. We may also observe a vector of baseline covariates unaffected by treatment, and all analysis in this section and the next is conditional on such variables.

Throughout, we will work with a non-atomic probability space $(\Omega, \Sigma, \mathrm{P})$, and all random variables in this paper are defined on this space. All inequalities and equalities involving density functions should be interpreted as holding μ-almost-everywhere. When I say "all" or "each" $y \in \mathcal{Y}$, I mean "μ-almost-all", and when I say that a property holds for "some $y \in \mathcal{Y}$", I mean that the set of $y \in \mathcal{Y}$ with the property has positive μ measure.

The basic setting is summarized in the following condition[3]:

Condition M (Generalized Roy Model)**.** *There exists random variables $\boldsymbol{V}$ and $\boldsymbol{\varepsilon}_Y$ defined on $(\Omega, \Sigma, \mathrm{P})$ that satisfy the two following properties:*

(i) *There exist measurable functions $g_D : \mathcal{Z} \times \text{supp}(\boldsymbol{V}) \to \mathcal{D}$ and $g_Y : \mathcal{D} \times \text{supp}(\boldsymbol{V}) \times \text{supp}(\boldsymbol{\varepsilon}_Y) \to \mathcal{Y}$ such that, almost surely,*

$$D = g_D(Z, \boldsymbol{V}) \,, \tag{1}$$

$$Y = g_Y(D, \boldsymbol{V}, \boldsymbol{\varepsilon}_Y) \,. \tag{2}$$

and, for each $d \in \mathcal{D}$, $g_Y(d, \mathbf{V}, \boldsymbol{\varepsilon}_Y)$ is μ-integrable and has a density function with respect to μ.

(ii) *The random variables $Z, \mathbf{V}, \boldsymbol{\varepsilon}_Y$ are mutually independent.*

Condition **M** introduces the unobservables $(\boldsymbol{V}, \boldsymbol{\varepsilon}_Y)$ and describes the causal relationship among the variables $(Y, D, Z, \boldsymbol{V}, \boldsymbol{\varepsilon}_Y)$. Condition **M**(i) makes explicit the structural equations (1)-(2) that are used below for formally defining and analyzing treatment selection models. Condition **M**(i) also imposes an integrability condition on g_Y to ensure that the relevant conditional expectations of Y exist. The exogeneity requirement on Z is captured by Condition **M**(ii), which implies that Z is randomly assigned. Condition **M**(ii) also imposes that the (potentially infinite-dimensional) unobservables $\boldsymbol{V}$ and $\boldsymbol{\varepsilon}_Y$ are independent from each other, which is an important feature for many of the results presented below. Informally, this assumption ensures that the residual uncertainty (i.e., after conditioning on $\boldsymbol{V}$) in the outcome is (allowed to be) rich enough so that assumptions on g_D and $\boldsymbol{V}$ alone cannot restrict the

[3]In Condition **M**, g_D and g_Y are deterministic functions. Some papers (e.g., Heckman and Pinto (2018)) model treatment selection as driven by a random function of Z and $\boldsymbol{V}$, and require the introduction of an independent shock $\boldsymbol{\varepsilon}_D$ to represent D as a deterministic function, $\tilde{g}_D(Z, \boldsymbol{V}, \boldsymbol{\varepsilon}_D)$; such models still satisfy Condition **M**, however, since we can just take $g_D(Z, \boldsymbol{V}) := \int_{\text{supp}(\boldsymbol{\varepsilon}_D)} \tilde{g}_D(Z, \boldsymbol{V}, \boldsymbol{e})\, d\mu(\boldsymbol{e})$.

outcome equation "too much". In particular, Condition **M**(ii) limits the role played by $\boldsymbol{V}$ in restricting the *joint* distribution of potential outcomes.

Remark 1. In the standard setting where Z is an excluded instrument, Condition **M** also implies an exclusion restriction. Since I allow Z (or some components of it) to be included in D, however, the results also apply to models where the conventional exclusion restriction may be violated (e.g., sample selection models). For an example of such an application, see Kroft, Mourifié, and Vayalinkal (2023).

The generalized Roy model described above is equivalent to a Neyman-Rubin potential outcomes model with random assignment of the instrument[4]. Indeed, potential outcomes can be defined, for $d \in \mathcal{D}$ and $z \in \mathcal{Z}$, as the random variables

$$D_z := g_D(z, \boldsymbol{V}) ,$$
$$Y_d := g_Y(d, \boldsymbol{V}, \boldsymbol{\varepsilon}_Y) .$$

In this notation,

$$D = \sum_{z \in \mathcal{Z}} \mathbb{1}\{Z = z\} D_z ,$$

$$Y = \sum_{d \in \mathcal{D}} \mathbb{1}\{D = d\} Y_d ,$$

and we have that Condition **M**(ii) implies $((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z})) \perp Z$, which is the usual random assignment assumption. I will use both notations, as convenient for the exposition.

Parent distributions. I now introduce some notions necessary for identification analysis. In the remainder, distributions of $((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z}), Z)$ will be called "parent" distributions and distributions of (Y, D, Z) will be called "observed data" distributions.

Denote by $\mathcal{M}$ the class of parent distributions that satisfy the above assumptions; i.e. all joint distributions of these variables that (i) have the correct supports and integrability properties and (ii) satisfy $((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z})) \perp Z$. In other words, a distribution Q is in $\mathcal{M}$ if and only if it witnesses Condition **M** for *some* distribution of (Y, D, Z) that satisfies our regularity conditions.

For any $Q \in \mathcal{M}$, denote by P_Q the distribution of the observed data under the parent distribution Q. Throughout the remainder, P will denote the (known) observed data distribution. Let $\mathcal{Q} \subseteq \mathcal{M}$ be the identified set for parent distributions under Condition **M**; formally, $\mathcal{Q} := \{Q \in \mathcal{M} : P_Q = P\}$, where the dependence of $\mathcal{Q}$ on P is suppressed.

Response-Types. I now introduce the notion of response-types, which is central to the remaining analysis. Define the (random) vector

$$\boldsymbol{T} := \boldsymbol{T}(\mathbf{V}) := (g_D(z, \boldsymbol{V}) : z \in \mathcal{Z}) = (D_z : z \in \mathcal{Z})$$

which denotes an individual's *response-type*. This is the same notion of response-types considered in Heckman and Pinto (2018).

Let $\boldsymbol{\mathcal{T}} := \operatorname{supp}(\boldsymbol{T}) = \{\boldsymbol{T}(\boldsymbol{v}) : \boldsymbol{v} \in \operatorname{supp}(\mathbf{V})\}$ denote the set of possible response-types. We can view $\boldsymbol{\mathcal{T}}$ as the set of possible functions from instruments to treatments, $\boldsymbol{\tau} : \mathcal{Z} \to \mathcal{D}$.

[4] I believe this is well-known. For completeness, I formally state and prove the equivalence of the two models in Appendix E.

For each response-type $\boldsymbol{\tau}$, denote by $\boldsymbol{\tau}(z)$ the treatment taken by those of response-type $\boldsymbol{\tau}$ when $Z = z$.

$\boldsymbol{\mathcal{T}}$ depends on the assumptions imposed on g_D and $\boldsymbol{V}$, and modulates the relationship between the observables and the unobservables. Indeed, Theorem T-1 in Heckman and Pinto (2018) implies that, for each treatment d and each outcome level y,

$$\sum_{\boldsymbol{\tau}\in\boldsymbol{\mathcal{T}}} \mathbb{1}\left\{\boldsymbol{\tau}(Z)=d\right\} \mathrm{P}\left(\boldsymbol{T}=\boldsymbol{\tau}\right) \mathrm{P}\left(Y_d \leq y \mid \boldsymbol{T}=\boldsymbol{\tau}\right) = \mathrm{P}\left(D=d \mid Z\right) \mathrm{P}\left(Y \leq y \mid D=d, Z\right) \ ,$$

almost surely.

If we do not place any restrictions on treatment selection, then $\boldsymbol{\mathcal{T}} = \mathcal{D}^{\mathcal{Z}}$. Such assumptions can also restrict the distribution of $\boldsymbol{T}$ in other ways, beyond just restricting its support. Importantly, I show below that assumptions on g_D and/or $\boldsymbol{V}$ have identifying power *only* through restrictions on the distribution of $\boldsymbol{T}$. The distribution of $\boldsymbol{T}$ is captured by the (unknown) vector of response-type probabilities, which we denote by $\boldsymbol{\pi}$,

$$\boldsymbol{\pi} := \left(\mathrm{P}\left(\boldsymbol{T}=\boldsymbol{\tau}\right) : \boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}}\right) .$$

Denote by $\boldsymbol{\pi}_Q$ the response-type probability vector induced by a particular parent distribution Q, i.e. $\boldsymbol{\pi}_Q := \left(Q\left(\boldsymbol{T}=\boldsymbol{\tau}\right) : \boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}}\right)$.

Remark 2. Response-types define a coarse partition of the support of $\mathbf{V}$:

$$\left\{\boldsymbol{v} \in \boldsymbol{T}^{-1}\left(\boldsymbol{\tau}\right)\right\} = \left\{\boldsymbol{v} \in \mathrm{supp}\left(\boldsymbol{V}\right) \mid \boldsymbol{T}\left(\boldsymbol{v}\right) = \boldsymbol{\tau}\right\} \ .$$

So, we can define a generalized Marginal Treatment Response (MTR) function based on $\mathbf{V}$,

$$MTR_d\left(\boldsymbol{v}\right) := E\left(Y_d \mid \boldsymbol{V}=\boldsymbol{v}\right) \ ,$$

and get that $E\left(Y_d \mid \boldsymbol{T}=\boldsymbol{\tau}\right) = \int_{\boldsymbol{T}^{-1}(\boldsymbol{\tau})} MTR_d\left(\boldsymbol{v}\right) \ d\mu(\boldsymbol{v})$.

In this draft, I focus on identification for counterfactual means and treatment effects conditional on response-types, and take the response-type support $\boldsymbol{\mathcal{T}}$ to be a set. The framework here can accommodate more general parameters if allow $\boldsymbol{\mathcal{T}}$ to be a multiset (i.e. allow "redundant" response-types). For instance, if we are interested in a parameter of the form $E\left(Y_d \mid \boldsymbol{V} \in \boldsymbol{\mathcal{V}}^*\right)$, for some measurable subset $\boldsymbol{\mathcal{V}}^*$, we can define response-types based on the possible values for the vector $\left(\left(g_D(z, \boldsymbol{V}) : z \in \mathcal{Z}\right), \mathbb{1}\left\{\boldsymbol{V} \in \boldsymbol{\mathcal{V}}^*\right\}\right)$. The remainder of the analysis goes through almost exactly in the same way[5], but otherwise "identical" response-types can be treated differently by the treatment selection model (formally defined below) specified by the analyst.

For (a simple) example, if we are interested in a hypothetical instrument value z^* that induces some members of certain response-types to switch from d to d' as the instrument switches from some particular z to z^*, we may wish to specify what proportion of each response-type will make the switch. I do not pursue such an approach further in this draft, but hope to include it in the future.

[5] Note that, for any particular $d \in \mathcal{D}$, $\mathbf{R}_d$ may have redundant columns even when $\boldsymbol{\mathcal{T}}$ has no redundant elements, and so, much of the analysis is not affected by the addition of more redundant columns.

2.2. **Identification in Treatment Selection Models.** To make progress, I first formalize the notion of "treatment selection model".

Definition 1 (Treatment Selection Model)**.** *A* ***treatment selection model*** *is a 3-tuple* $\mathbb{S} = (\mathcal{G}_{\mathbb{S}}, \boldsymbol{\mathcal{V}}_{\mathbb{S}}, \mathcal{F}_{\mathbb{S}})$ *such that*

$$\begin{aligned} \boldsymbol{\mathcal{V}}_{\mathbb{S}} &\text{ is a set },\\ \mathcal{G}_{\mathbb{S}} &\subseteq \{\mathcal{Z} \times \boldsymbol{\mathcal{V}}_{\mathbb{S}} \to \mathcal{D}\} ,\\ \text{and } \mathcal{F}_{\mathbb{S}} &\subseteq \{F : F \text{ is a distribution with } \operatorname{supp}(F) \subseteq \boldsymbol{\mathcal{V}}_{\mathbb{S}}\} . \end{aligned}$$

A parent distribution $Q \in \mathcal{M}$ *is said to satisfy treatment selection model* $\mathbb{S}$ *if and only if there exists random variables* $(Y, D, Z, \boldsymbol{V}, \varepsilon_Y)$ *and functions* g_D, g_Y *such that they satisfy Condition* ***M*** *and have* $\operatorname{Law}((g_Y(d, \boldsymbol{V}, \varepsilon_Y) : d \in \mathcal{D}), (g_D(z, \boldsymbol{V}) : z \in \mathcal{Z}), Z) = Q$*, with* $\operatorname{supp}(\boldsymbol{V}) \subseteq \boldsymbol{\mathcal{V}}_{\mathbb{S}}$*,* $\operatorname{Law}(\boldsymbol{V}) \in \mathcal{F}_{\mathbb{S}}$ *and* $g_D \in \mathcal{G}_{\mathbb{S}}$*.*

Remark 3. Although the formal definition above takes treatment selection models to be specified as a 3-tuple, in practice we may often only impose assumptions on g_D (e.g., shape restrictions), leaving the distribution and support of $\boldsymbol{V}$ unrestricted.

For any treatment selection model $\mathbb{S}$, define $\mathcal{Q}_{\mathbb{S}}$ as the identified set of parent distributions that satisfy $\mathbb{S}$,

$$\mathcal{Q}_{\mathbb{S}} := \{Q \in \mathcal{Q} \mid Q \text{ satisfies } \mathbb{S}\} .$$

Next, define $\mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ as the *a priori* restrictions imposed on the distribution of $\boldsymbol{T}$, represented by the vector response-type probabilities $\boldsymbol{\pi}$, i.e.

$$\mathcal{F}_{\boldsymbol{T}}(\mathbb{S}) := \{\boldsymbol{\pi}_Q \mid Q \in \mathcal{M} \text{ and } Q \text{ satisfies } \mathbb{S}\} .$$

$\mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ is a feature of the treatment selection model $\mathbb{S}$ and does not depend on the observed data distribution P. Moreover, $\mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ can be represented as a subset of the set of all probability mass functions on $\mathcal{D}^{\mathcal{Z}}$ (i.e. the standard $\mathcal{D}^{\mathcal{Z}}$-simplex, $\Delta^{\mathcal{D}^{\mathcal{Z}}}$), and is therefore always a finite-dimensional bounded set. We will refer to the set $\mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ as the *type restriction* associated with selection model $\mathbb{S}$.

As promised earlier, the following result shows that treatment selection models have identifying power only through their associated type restriction.

Theorem 1. *For any treatment selection model* $\mathbb{S}$*,*

$$\mathcal{Q}_{\mathbb{S}} = \{Q \in \mathcal{Q} \mid \boldsymbol{\pi}_Q \in \mathcal{F}_{\boldsymbol{T}}(\mathbb{S})\} .$$

Proof. See Appendix A.1. □

Finally, I introduce some general notions, used in the remainder of the paper. Define $\boldsymbol{\mathcal{T}}(\mathbb{S})$ as the set of all response-types not *a priori* eliminated by the selection model,

$$\begin{aligned} \boldsymbol{\mathcal{T}}(\mathbb{S}) &:= \left\{\boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}} \;\middle|\; \boldsymbol{\pi} \in \mathcal{F}_{\boldsymbol{T}}(\mathbb{S}) \text{ such that } \boldsymbol{\pi}_{(\boldsymbol{\tau})} > 0\right\} \\ &= \left\{\boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}} \;\middle|\; \exists Q \in \mathcal{Q}_{\mathbb{S}} \text{ such that } Q(\boldsymbol{T} = \boldsymbol{\tau}) > 0\right\} , \end{aligned}$$

and note that $\mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ can always be represented as a subset of $\mathbb{R}^{|\boldsymbol{\mathcal{T}}(\mathbb{S})|}$.

For any subset $\boldsymbol{\mathcal{T}} \subseteq \mathcal{D}^{\mathcal{Z}}$, define $\mathcal{Q}_{\boldsymbol{\mathcal{T}}}$ as the identified set for parent distributions with response-type support in $\boldsymbol{\mathcal{T}}$,

$$\mathcal{Q}_{\boldsymbol{\mathcal{T}}} := \left\{ Q \in \mathcal{Q} \;\middle|\; \sum_{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}} Q\left(\boldsymbol{T} = \boldsymbol{\tau}\right) = 1 \right\} .$$

In general, $\mathcal{Q}_{\mathbb{S}} \subseteq \mathcal{Q}_{\boldsymbol{\mathcal{T}}(\mathbb{S})}$, but in the important special case where $\mathbb{S}$ simply eliminates some response-types, we have that $\mathcal{Q}_{\mathbb{S}} = \mathcal{Q}_{\boldsymbol{\mathcal{T}}(\mathbb{S})}$.

Remark 4 ("Translating" a Selection Model into a Type Restriction)**.** I will generally treat $\mathcal{F}_{\boldsymbol{T}}\left(\mathbb{S}\right)$ as known, given $\mathbb{S}$. I emphasize, however, that "translating" such models from more "structural" representations can be non-trivial, in general. A well-known example of such translation is Vytlacil (2002, 2006).

Given some representation of $\mathbb{S}$, it is often much easier to derive $\boldsymbol{\mathcal{T}}\left(\mathbb{S}\right)$ than $\mathcal{F}_{\boldsymbol{T}}\left(\mathbb{S}\right)$. In cases where $\mathcal{F}_{\boldsymbol{T}}\left(\mathbb{S}\right)$ is not known but $\boldsymbol{\mathcal{T}}\left(\mathbb{S}\right)$ is, this paper's framework can be used to construct identified sets for parameters over $Q \in \mathcal{Q}_{\boldsymbol{\mathcal{T}}(\mathbb{S})}$ to obtain valid bounds (potentially not sharp), useful for (potentially conservative) inference.

The next section develops methodology for identifying response-type conditional counterfactual distributions and parameters.

3. IDENTIFICATION OF COUNTERFACTUALS

I introduce the following shorthand notation, used throughout the remainder. For each $Q \in \mathcal{M}$, define

$$\boldsymbol{\pi\xi}_{Y_d,Q}\left(y\right) := \left(Q\left(\boldsymbol{T} = \boldsymbol{\tau}\right) f_{Q;Y_d}\left(y \mid \boldsymbol{T} = \boldsymbol{\tau}\right) : \boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}} \right) ,$$

where $f_{Q;Y_d}\left(y \mid \boldsymbol{T} = \boldsymbol{\tau}\right)$ is the density associated with $Q\left(Y_d \leq y \mid \boldsymbol{T} = \boldsymbol{\tau}\right)$. Note that $\int_{\mathcal{Y}} \boldsymbol{\pi\xi}_{Y_d,Q}\left(y\right)\, d\mu\left(y\right) = \boldsymbol{\pi}_Q$.

When the response-type support $\boldsymbol{\mathcal{T}}$ is clear, we will treat $\boldsymbol{\pi}_Q$ and $\boldsymbol{\pi\xi}_{Y_d,Q}(y)$ as $|\boldsymbol{\mathcal{T}}|$-vectors (instead of $|\mathcal{D}^{\mathcal{Z}}|$-vectors), a slight abuse of notation. We will often suppress the dependence on Q and just write $\boldsymbol{\pi}$ and $\boldsymbol{\pi\xi}_{Y_d}(y)$.

For any $\boldsymbol{\mathcal{T}} \subseteq \mathcal{D}^{\mathcal{Z}}$, let $\boldsymbol{\mathcal{T}} =: \left\{\boldsymbol{\tau}_1, \ldots, \boldsymbol{\tau}_{|\boldsymbol{\mathcal{T}}|}\right\}$ be an ordering of response-types, and let $\mathcal{Z} =: \left\{z_1, \ldots z_{|\mathcal{Z}|}\right\}$ be some ordering of the instrument levels; throughout, assume all relevant tuples follow this ordering.

Next, define, for each $d \in \mathcal{D}$, the $|\mathcal{Z}| \times |\boldsymbol{\mathcal{T}}|$ matrix $\mathbf{R}_{d,\boldsymbol{\mathcal{T}}}$ as the binary matrix such that its (i,j)-th component is $\mathbb{1}\left\{\boldsymbol{\tau}_j\left(z_i\right) = d\right\}$. Note that $\mathbf{R}_{d,\boldsymbol{\mathcal{T}}}$ is feature of the model itself: it is known from $\boldsymbol{\mathcal{T}}$, and does not depend on the observed data. When the $\boldsymbol{\mathcal{T}}$ of interest is clear, we will often suppress the dependence on $\boldsymbol{\mathcal{T}}$ and just write $\mathbf{R}_d$[6].

Finally, define $\mathbf{pf}_d$ as the (observed) vector of propensity-score-weighted densities,

$$\mathbf{pf}_d\left(y\right) := \left(P\left(D = d \mid Z = z\right) f_{P;Y}\left(y \mid D = d, Z = z\right) : z \in \mathcal{Z} \right) ,$$

where $f_{P;Y}\left(y \mid D = d, Z = z\right)$ is the density associated with $P\left(Y \leq y \mid D = d, Z = z\right)$.

The following theorem characterizes the role of $\boldsymbol{\mathcal{T}}$ in modulating the relationship between the observed data distribution P and the set of possible parent distributions.

[6] Heckman and Pinto (2018) denote this matrix $\boldsymbol{B}_d$, in their notation.

Theorem 2 (Sharp Characterization of $\mathcal{Q}_{\mathcal{T}}$). *For any $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$, $Q \in \mathcal{Q}_{\mathcal{T}}$ if and only if $Q \in \mathcal{M}$, $Q_Z = P_Z$ and, for each $d \in \mathcal{D}$ and $y \in \mathcal{Y}$, the following equality holds*

$$\mathbf{R}_d \boldsymbol{\pi} \boldsymbol{\xi}_{Y_d}(y) = \mathbf{pf}_d(y) \ . \tag{$\star$}$$

Proof. See Appendix A.2. □

As noted earlier, the $\Longrightarrow$ direction in Theorem 2 follows immediately from Theorem T-1 in Heckman and Pinto (2018). The interesting part of this result is that it is sharp (i.e. the $\Longleftarrow$ part of the above result), but it is not immediately operationalizable since (in general) it represents a continuum of equalities. Nevertheless, Theorem 2 suggests taking $\boldsymbol{\pi}\boldsymbol{\xi}_{Y_d,Q}$, for $d \in \mathcal{D}$, as the primitives in our model. In other words, this result suggests that the key ingredients for identification analysis in this setting are the distributions $(Y_d, \boldsymbol{T})$, for $d \in \mathcal{D}$.

Remark 5. If $\mathcal{Y}$ is finite, then we can use Theorem 2 to set up our identification problem as one large linear programming problem and proceed using, e.g, Fang, Santos, Shaikh, and Torgovitsky (2023) or Cho and Russell (2023) for inference. I do not seriously pursue this approach in this draft. The main focus of this paper is on an alternate (arguably less "black box") approach that can accommodate both finite and continuous outcomes; in many cases, linear programming will still play a role in computation, see Section 5.

3.1. Identified Set for Response-Type Probabilities and Mean Counterfactuals.

In this subsection, we take $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$ as given. Define $\Pi_{\mathcal{T}}$ as the identified set for $\boldsymbol{\pi}$ when the response-type support is restricted to $\mathcal{T}$,

$$\Pi_{\mathcal{T}} := \{\boldsymbol{\pi}_Q : Q \in \mathcal{Q}_{\mathcal{T}}\} \ ,$$

which we will generally view as a subset of $\mathbb{R}^{|\mathcal{T}|}$.

Before proceeding, we define what it means for a set to be a convex polytope. This and related notions are used throughout the remainder of the paper.

Definition 2. *A set $K \subseteq \mathbb{R}^n$ is a* ***convex polytope*** *if and only if there exists, for some $m \in \mathbb{N}$, a matrix $A \in \mathbb{R}^{m\times n}$ and vector $b \in \mathbb{R}^m$ such that*

$$K = \{x \in \mathbb{R}^n \mid Ax \le b\} \ .$$

If K is a bounded convex polytope, then K is the convex hull of a finite set in $\mathbb{R}^n$.

Fix some $d \in \mathcal{D}$. For any $y \in \mathcal{Y}$, let $\mathcal{B}_d(\mathbf{pf}_d(y)) \subseteq \mathbb{R}^{|\mathcal{T}|}_{\ge 0}$ be the (potentially unbounded) convex polytope defined by the set of non-negative solutions to ($\star$) at (d, y),

$$\mathcal{B}_d(\mathbf{pf}_d(y)) := \left\{\mathbf{x} \in \mathbb{R}^{|\mathcal{T}|}_{\ge 0} \,\middle|\, \mathbf{R}_d \mathbf{x} = \mathbf{pf}_d(y)\right\} \ .$$

$\mathcal{B}_d(\mathbf{pf}_d(y))$ is identified since $\mathbf{pf}_d$ depends only on the observed data distribution P.

Next, define $\boldsymbol{\Gamma}_d(\mathcal{T})$ as the set of non-negative measurable vector-valued functions $\boldsymbol{g} : \mathcal{Y} \to \mathbb{R}^{|\mathcal{T}|}_{\ge 0}$ such that $\boldsymbol{g}(y) \in \mathcal{B}_d(\mathbf{pf}_d(y))$ for each $y \in \mathcal{Y}$,[7]

$$\boldsymbol{\Gamma}_d(\mathcal{T}) := \left\{\boldsymbol{g} \in \left\{\mathcal{Y} \to \mathbb{R}^{|\mathcal{T}|}_{\ge 0}\right\} \,\middle|\, \begin{array}{l} \boldsymbol{g} \text{ is measurable,} \\ \boldsymbol{g}(y) \in \mathcal{B}_d(\mathbf{pf}_d(y)) \, \forall y \in \mathcal{Y} \end{array}\right\} \ .$$

[7]Here, the correspondence $y \mapsto \mathcal{B}_d(\mathbf{pf}_d(y))$ can be viewed as a (generalized) *polytope bundle* (Billera and Sturmfels, 1992). In the language of fibre bundles, each $\mathcal{B}_d(\mathbf{pf}_d(y))$ is a *fibre* of the bundle $y \mapsto \mathcal{B}_d(\mathbf{pf}_d(y))$, and each $\boldsymbol{g} \in \boldsymbol{\Gamma}_d$ is a *section* of this bundle.

We will generally suppress the dependence of $\mathbf{\Gamma}_d$ on $\boldsymbol{\mathcal{T}}$. Note that if $\boldsymbol{\mathcal{T}}$ is such that $\mathcal{B}_d\left(\mathbf{pf}_d\left(y\right)\right)=\emptyset$ for some $y\in\mathcal{Y}$, then $\mathbf{\Gamma}_d=\emptyset$. Theorem 2 implies that $\boldsymbol{\pi}\boldsymbol{\xi}_{Y_d}$ satisfies ($\star$) for every $y\in\mathcal{Y}$, and so, we must have that $\boldsymbol{\pi}\boldsymbol{\xi}_{Y_d}\in\mathbf{\Gamma}_d$.

Next, consider the set, denoted $\Pi_{\boldsymbol{\mathcal{T}}|d}$, formed by the integrals of functions in $\mathbf{\Gamma}_d$[8],

$$\Pi_{\boldsymbol{\mathcal{T}}|d}:=\left\{\int_{\mathcal{Y}}\boldsymbol{g}(y)\ d\mu\left(y\right)\ \middle|\ \boldsymbol{g}\in\mathbf{\Gamma}_d\right\}\ .$$

Clearly, $\boldsymbol{\pi}\boldsymbol{\xi}_{Y_d}\in\mathbf{\Gamma}_d$ implies that $\boldsymbol{\pi}\in\Pi_{\boldsymbol{\mathcal{T}}|d}$. By linearity of integration, we have that $\Pi_{\boldsymbol{\mathcal{T}}|d}$ is a closed convex set, and so, we will work with its *support function*, which we define next.

Definition 3. *For any non-empty closed convex set $C\subseteq\mathbb{R}^n$, let $h_C:\mathbb{R}^n\to\mathbb{R}$ be its **support function**, which is defined as the convex function such that, for every $\boldsymbol{c}\in\mathbb{R}^n$,*

$$h_C\left(\boldsymbol{c}\right):=\sup_{\boldsymbol{x}\in C}\boldsymbol{c}^{\intercal}\boldsymbol{x}\ .$$

Every non-empty closed convex set is uniquely determined by its support function.

Whenever $\mathcal{B}_d\left(\mathbf{pf}_d\left(y\right)\right)\neq\emptyset$, denote the support function of $\mathcal{B}_d\left(\mathbf{pf}_d\left(y\right)\right)$ as $h_{\mathbf{R}_d}\left(\,\cdot\,;\ \mathbf{pf}_d\left(y\right)\right)$,

$$h_{\mathbf{R}_d}\left(\boldsymbol{c};\ \mathbf{pf}_d\left(y\right)\right):=h_{\mathcal{B}_d\left(\mathbf{pf}_d\left(y\right)\right)}\left(\boldsymbol{c}\right)=\sup_{\boldsymbol{x}\geq 0:\mathbf{R}_d\boldsymbol{x}=\mathbf{pf}_d(y)}\boldsymbol{c}^{\intercal}\boldsymbol{x}\ .$$

In Section 5, I discuss the computation and estimation of the function $h_{\mathbf{R}_d}$, which is an important object in our analysis. The next result relates $h_{\mathbf{R}_d}$ to the support function of $\Pi_{\boldsymbol{\mathcal{T}}|d}$.

Lemma 1. *If $\boldsymbol{\mathcal{T}}\subseteq\mathcal{D}^{\mathcal{Z}}$ is such that $\mathcal{B}_d\left(\mathbf{pf}_d\left(y\right)\right)\neq\emptyset$ for all $y\in\mathcal{Y}$, then, for all $\boldsymbol{c}\in\mathbb{R}^{|\boldsymbol{\mathcal{T}}|}$*

$$h_{\Pi_{\boldsymbol{\mathcal{T}}|d}}\left(\boldsymbol{c}\right)=\int_{\mathcal{Y}}h_{\mathbf{R}_d}\left(\boldsymbol{c};\ \mathbf{pf}_d\left(y\right)\right)\ d\mu\left(y\right)\ .$$

Further, $\Pi_{\boldsymbol{\mathcal{T}}|d}$ is a convex polytope.

Proof. See Appendix A.4. □

Finally, the argument that $\boldsymbol{\pi}\in\Pi_{\boldsymbol{\mathcal{T}}|d}$ holds for each $d\in\mathcal{D}$, and so, we must have $\boldsymbol{\pi}\in\bigcap_{d\in\mathcal{D}}\Pi_{\boldsymbol{\mathcal{T}}|d}$. The next result shows that this is sharp.

Theorem 3. *For any $\boldsymbol{\mathcal{T}}\subseteq\mathcal{D}^{\mathcal{Z}}$,*

$$\Pi_{\boldsymbol{\mathcal{T}}}=\bigcap_{d\in\mathcal{D}}\Pi_{\boldsymbol{\mathcal{T}}|d}\ ,$$

and $\Pi_{\boldsymbol{\mathcal{T}}}$ is a compact convex polytope in $\mathbb{R}^{|\boldsymbol{\mathcal{T}}|}_{\geq 0}$. Moreover, $\mathcal{Q}_{\boldsymbol{\mathcal{T}}}\neq\emptyset$ if and only if $\Pi_{\boldsymbol{\mathcal{T}}}\neq\emptyset$.

Proof. See Appendix A.3 □

[8] $\Pi_{\boldsymbol{\mathcal{T}}|d}$ is the *Minkowski integral* of the bundle $y\mapsto\mathcal{B}_d\left(\mathbf{pf}_d\left(y\right)\right)$, which can be viewed as the continuous analogue of the Minkowski sum. Indeed, if $\mathcal{Y}$ is finite, then $\Pi_{\boldsymbol{\mathcal{T}}|d}$ is just the Minkowski sum of the convex polytopes $\left\{\mathcal{B}_d\left(\mathbf{pf}_d\left(y\right)\right)\right\}_{y\in\mathcal{Y}}$.

Remark 6. Theorem 3 does not assume that Condition **M** holds (i.e., that $\mathcal{Q} \neq \emptyset$); indeed, setting $\boldsymbol{\mathcal{T}} = \mathcal{D}^{\mathcal{Z}}$ in Theorem 3 yields (sharp) testable implications for Condition **M** (i.e. a test for $\mathcal{Q} \neq \emptyset$). If we assume that Condition **M**(i) holds, then this represents a test for random assignment of Z.

Moreover, Theorem 3 immediately implies the following general result.

Corollary 1. *For any treatment selection model $\mathbb{S}$, define the identified set for response-type probabilities under $\mathbb{S}$ as $\Pi_{\mathbb{S}} := \{\boldsymbol{\pi}_Q : Q \in \mathcal{Q}_{\mathbb{S}}\}$. Then,*

$$\Pi_{\mathbb{S}} := \Pi_{\boldsymbol{\mathcal{T}}(\mathbb{S})} \cap \mathcal{F}_{\boldsymbol{T}}(\mathbb{S}) \ ,$$

where $\Pi_{\mathbb{S}}$ and $\mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ are represented as subsets of $\mathbb{R}^{|\boldsymbol{\mathcal{T}}(\mathbb{S})|}$. Moreover, if $\mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ is a convex polytope, then $\Pi_{\mathbb{S}}$ is a convex polytope.

Proof. The first claim follows immediately from Theorems 1 and 3. The second claim follows from Theorem 3 by noting that the intersection of convex polytopes is always a convex polytope. □

Remark 7. We have that $\boldsymbol{\Gamma}_d$, which only depends on $\mathbb{S}$ through $\boldsymbol{\mathcal{T}}(\mathbb{S})$, represents pointwise restrictions on $\boldsymbol{\pi\xi}_{Y_d}$; informally, this can interpreted as the "direct" restrictions imposed on the distribution $(Y_d, \boldsymbol{T})$. For each $d \in \mathcal{D}$, these direct restrictions on $(Y_d, \boldsymbol{T})$, in turn, impose restrictions on the distribution of $\boldsymbol{T}$, and these restrictions must all be compatible with each other and also with any additional restrictions imposed by $\mathbb{S}$.

The "direct" restrictions on $(Y_d, \boldsymbol{T})$ can be viewed as restricting both the conditional distribution $Y_d|\boldsymbol{T}$ and the *relationship* between the distribution of $Y_d|\boldsymbol{T}$ and the distribution of $\boldsymbol{T}$, and then the "compatibility" condition restricts the distribution of $\boldsymbol{T}$ which, in turn, restricts the distribution of $Y_d|\boldsymbol{T}$ only indirectly via the restricted relationship between the distributions of $Y_d|\boldsymbol{T}$ and $\boldsymbol{T}$.

In particular, the set of allowed response-types directly restricts the distributions of $Y_d|\boldsymbol{T}$ while any other restrictions on the distribution of $\boldsymbol{T}$ only restricts "how much" mass is available to allocate to $Y_d|\boldsymbol{T}$ but not "where" (in $\mathcal{Y}$) the available mass can be allocated. This is intuitively similar to Horowitz and Manski (1995)'s seminal result on the partial identification of mixture components.

I emphasize, however, that these "indirect" restrictions can nevertheless provide significant identification power in many cases, which is somewhat different from simpler settings (e.g., binary Z and D). For an extreme case, see Remark 9 which outlines knife-edge conditions under which indirect restrictions can deliver point-identification.

As a by-product of the proof of Theorem 3, we have the following corollary, which also helps highlight the intuition discussed above.

Corollary 2 (Identified Set for Marginal Outcome Distributions)**.** *For any treatment selection model $\mathbb{S}$, we have that the identified set for $\boldsymbol{\pi\xi}_{Y_d}$ under $\mathbb{S}$ is given by*

$$\left\{\boldsymbol{\pi\xi}_{Y_d,Q} \;\middle|\; Q \in \mathcal{Q}_{\mathbb{S}}\right\} = \left\{\boldsymbol{g} \in \boldsymbol{\Gamma}_d(\boldsymbol{\mathcal{T}}(\mathbb{S})) \;\middle|\; \int_{\mathcal{Y}} \boldsymbol{g}(y)\ d\mu(y) \in \Pi_{\mathbb{S}}\right\} .$$

Further, the joint identified region for $\left(\boldsymbol{\pi}\boldsymbol{\xi}_{Y_d} : d \in \mathcal{D}\right)$ *under* $\mathbb{S}$ *is given by*

$$\begin{aligned}&\left\{\left(\boldsymbol{\pi}\boldsymbol{\xi}_{Y_d,Q} : d \in \mathcal{D}\right) \,\middle|\, Q \in \mathcal{Q}_{\mathbb{S}}\right\} \\ &= \left\{(\boldsymbol{g}_d : d \in \mathcal{D}) \in \prod_{d\in\mathcal{D}} \boldsymbol{\Gamma}_d\left(\boldsymbol{\mathcal{T}}(\mathbb{S})\right) \,\middle|\, \exists \boldsymbol{\pi} \in \Pi_{\mathbb{S}} \text{ s.t. } \int_{\mathcal{Y}} \boldsymbol{g}_d(y)\ d\mu(y) = \boldsymbol{\pi}\ \forall d \in \mathcal{D}\right\} .\end{aligned}$$

Proof. Follows from the proof of Theorem 3. See Appendix A.3 for details. □

Corollary 2 can be applied to obtain sharp bounds on mean outcomes and treatment effects, as follows. Suppose we are interested in the identified set for $E\left(\kappa(Y_d) \mid \boldsymbol{T} \subseteq \boldsymbol{\mathcal{T}}^*\right)$, where the function $\kappa : \mathcal{Y} \to \mathbb{R}$ and subset $\boldsymbol{\mathcal{T}}^* \subseteq \boldsymbol{\mathcal{T}}$ are known. Note that

$$E\left[\kappa(Y_d) \mid \boldsymbol{T} \subseteq \boldsymbol{\mathcal{T}}^*\right] = \frac{\sum_{\boldsymbol{\tau}\in\boldsymbol{\mathcal{T}}^*} \mathrm{P}(\boldsymbol{T} = \boldsymbol{\tau})\, E\left(\kappa(Y_d) \mid \boldsymbol{T} = \boldsymbol{\tau}\right)}{\sum_{\boldsymbol{\tau}\in\boldsymbol{\mathcal{T}}^*} \mathrm{P}(\boldsymbol{T} = \boldsymbol{\tau})} .$$

Corollary 2 suggests a representation for the identified set for parameters of this form. To proceed, first define the following sets, for any $\boldsymbol{\mathcal{T}} \subseteq \mathcal{D}^{\mathcal{Z}}, d \in \mathcal{D}$, and selection model $\mathbb{S}$,

$$\begin{aligned}\Theta_{\boldsymbol{\mathcal{T}}|d}(\kappa) &:= \left\{\left(\int_{\mathcal{Y}} \kappa(y)\, \boldsymbol{g}(y)\ d\mu(y), \int_{\mathcal{Y}} \boldsymbol{g}(y)\ d\mu(y)\right) \in \mathbb{R}^{|\boldsymbol{\mathcal{T}}|} \times \mathbb{R}^{|\boldsymbol{\mathcal{T}}|} \,\middle|\, \boldsymbol{g} \in \boldsymbol{\Gamma}_d(\boldsymbol{\mathcal{T}})\right\} , \\ \tilde{\Pi}_{\mathbb{S}} &:= \left\{(\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \mathbb{R}^{|\boldsymbol{\mathcal{T}}|} \times \mathbb{R}^{|\boldsymbol{\mathcal{T}}|} \,\middle|\, \boldsymbol{\theta}_d \in \kappa(\mathcal{Y})^{|\boldsymbol{\mathcal{T}}|},\ \boldsymbol{\pi} \in \Pi_{\mathbb{S}}\right\} .\end{aligned}$$

The set $\tilde{\Pi}_{\mathbb{S}}$ is just the representation of $\Pi_{\mathbb{S}}$ in a higher dimensional space. $\Theta_{\boldsymbol{\mathcal{T}}|d}$ is a closed convex set and, as with $\Pi_{\boldsymbol{\mathcal{T}}|d}$, the support function of $\Theta_{\boldsymbol{\mathcal{T}}|d}$ can be obtained in terms of $h_{\mathbf{R}_d}$, as follows.

Lemma 2. *If* $\boldsymbol{\mathcal{T}} \subseteq \mathcal{D}^{\mathcal{Z}}$ *is such that* $\mathcal{B}_d(\mathbf{pf}_d(y)) \neq \emptyset$ *for all* $y \in \mathcal{Y}$*, then, for all* $(\boldsymbol{c}_1, \boldsymbol{c}_2) \in \mathbb{R}^{|\boldsymbol{\mathcal{T}}|} \times \mathbb{R}^{|\boldsymbol{\mathcal{T}}|}$,

$$h_{\Theta_{\boldsymbol{\mathcal{T}}|d}(\kappa)}(\boldsymbol{c}_1, \boldsymbol{c}_2) = \int_{\mathcal{Y}} h_{\mathbf{R}_d}\left(\kappa(y)\, \boldsymbol{c}_1 + \boldsymbol{c}_2\ ;\ \mathbf{pf}_d(y)\right) d\mu(y) .$$

Proof. See Appendix A.4. □

We are now ready to state the following identification result for mean counterfactual outcomes and treatment effects, as suggested by Corollary 2.

Theorem 4 (Identified Set for Mean Counterfactuals)**.** *Let a treatment selection model* $\mathbb{S}$*, a function* $\kappa : \mathcal{Y} \to \mathbb{R}$*, and a subset* $\boldsymbol{\mathcal{T}}^* \subseteq \boldsymbol{\mathcal{T}}(\mathbb{S})$ *be given, and let*

$$\boldsymbol{c} := \left(\mathbb{1}\left\{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}^*\right\} : \boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}(\mathbb{S})\right) .$$

Then, for any $d \in \mathcal{D}$,

$$\begin{aligned}\left\{Q(\boldsymbol{T} \in \boldsymbol{\mathcal{T}}^*)\, E_Q\left[\kappa(Y_d) \mid \boldsymbol{T} \in \boldsymbol{\mathcal{T}}^*\right] \,\middle|\, Q \in \mathcal{Q}_{\mathbb{S}}\right\} &= \left\{\boldsymbol{c}^{\intercal}\boldsymbol{\theta}_d \,\middle|\, (\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}}\right\} , \\ \textit{and } \left\{E_Q\left[\kappa(Y_d) \mid \boldsymbol{T} \in \boldsymbol{\mathcal{T}}^*\right] \,\middle|\, Q \in \mathcal{Q}_{\mathbb{S}}\right\} &= \left\{\frac{\boldsymbol{c}^{\intercal}\boldsymbol{\theta}_d}{\boldsymbol{c}^{\intercal}\boldsymbol{\pi}} \,\middle|\, (\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}}\right\} .\end{aligned}$$

Further, for any $d \neq d' \in \mathcal{D}$,

$$\begin{aligned}&\left\{Q(\boldsymbol{T} \in \boldsymbol{\mathcal{T}}^*)\, E_Q\left[\kappa(Y_d) - \kappa(Y_{d'}) \mid \boldsymbol{T} \in \boldsymbol{\mathcal{T}}^*\right] \,\middle|\, Q \in \mathcal{Q}_{\mathbb{S}}\right\} \\ &\qquad = \left\{\boldsymbol{c}^{\intercal}\boldsymbol{\theta}_d - \boldsymbol{c}^{\intercal}\boldsymbol{\theta}_{d'} \,\middle|\, \begin{array}{l}(\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}} , \\ (\boldsymbol{\theta}_{d'}, \boldsymbol{\pi}) \in \Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d'}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}}\end{array}\right\} ,\end{aligned}$$

and

$$\left\{E_Q\left[\kappa\left(Y_d\right)-\kappa\left(Y_{d'}\right)\;\middle|\;\boldsymbol{T}\in\boldsymbol{\mathcal{T}}^*\right]\;\middle|\;Q\in\mathcal{Q}_{\mathbb{S}}\right\}$$

$$=\left\{\frac{\boldsymbol{c}^{\intercal}\boldsymbol{\theta}_d-\boldsymbol{c}^{\intercal}\boldsymbol{\theta}_{d'}}{\boldsymbol{c}^{\intercal}\boldsymbol{\pi}}\;\middle|\;\begin{array}{l}\left(\boldsymbol{\theta}_d,\boldsymbol{\pi}\right)\in\Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}\left(\kappa\right)\cap\tilde{\Pi}_{\mathbb{S}}\;,\\ \left(\boldsymbol{\theta}_{d'},\boldsymbol{\pi}\right)\in\Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d'}\left(\kappa\right)\cap\tilde{\Pi}_{\mathbb{S}}\end{array}\right\}.$$

Proof. Follows from the proof of Theorem 3, via Corollary 2. See Appendix A.3. □

Remark 8. For simplicity, Theorem 4 only reports the case where $\boldsymbol{c}$ is a binary vector and only considers treatment effects involving two identically weighted outcomes, but the analogous result applies to any weighted average of response-type conditional outcome means.

Moreover, note that additional assumptions on the response-type conditional means can be imposed by adding suitable conditions to the definition of $\tilde{\Pi}_{\mathbb{S}}$. This includes assumptions like $E\left(Y_d \mid \boldsymbol{T}=\boldsymbol{\tau}\right)\geq E\left(Y_{d'}\mid\boldsymbol{T}=\boldsymbol{\tau}\right)$, or (discrete analogues to) the "comparable compliers" assumption of Mountjoy (2022), or more complex linear relationships. Non-linear restrictions can also be imposed, but this will complicate the computation of the support function of $\tilde{\Pi}_{\mathbb{S}}$, required for the inference approach proposed in this paper (see Section 4).

3.1.1. *Point Identification.* For completeness, I also revisit sufficient conditions for point identification, due to Heckman and Pinto (2018).

Proposition 5 (Heckman and Pinto (2018))**.** *For any treatment selection model* $\mathbb{S}$ *with* $\mathcal{Q}_{\mathbb{S}}\neq\emptyset$*, the following statements hold, under* $\mathbb{S}$*:*

(i) *If* $\boldsymbol{c}\in\text{row-span}\left(\mathbf{R}_{d,\boldsymbol{\mathcal{T}}(\mathbb{S})}\right)$*, then* $\boldsymbol{c}^{\intercal}\boldsymbol{\pi}\boldsymbol{\xi}_{Y_d}$ *is point-identified (up to a* μ*-null subset of* $\mathcal{Y}$*)*

(ii) *If* $\boldsymbol{c}\in\text{row-span}\left(\begin{bmatrix}\mathbf{R}_{1,\boldsymbol{\mathcal{T}}(\mathbb{S})}\\ \vdots\\ \mathbf{R}_{N_D,\boldsymbol{\mathcal{T}}(\mathbb{S})}\end{bmatrix}\right)$*, then* $\boldsymbol{c}^{\intercal}\boldsymbol{\pi}$ *is point-identified*

Proposition 5 is mathematically equivalent to Theorem T-2 in Heckman and Pinto (2018), where the result is stated in terms of Moore-Penrose inverses of $\mathbf{R}_d$. Note that Proposition 5(i) is also sufficient for the point identification of mean outcomes, since $\boldsymbol{c}\in\text{row-span}\left(\mathbf{R}_{d,\boldsymbol{\mathcal{T}}(\mathbb{S})}\right)$ implies that $\left\{\boldsymbol{c}^{\intercal}\boldsymbol{\theta}_d \;\middle|\; \left(\boldsymbol{\theta}_d,\boldsymbol{\pi}\right)\in\Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}\left(\kappa\right)\cap\tilde{\Pi}_{\mathbb{S}}\right\}$ and $\left\{\frac{\boldsymbol{c}^{\intercal}\boldsymbol{\theta}_d}{\boldsymbol{c}^{\intercal}\boldsymbol{\pi}} \;\middle|\; \left(\boldsymbol{\theta}_d,\boldsymbol{\pi}\right)\in\Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}\left(\kappa\right)\cap\tilde{\Pi}_{\mathbb{S}}\right\}$ are singletons. Similarly, for point identification of treatment effects, $\boldsymbol{c}\in\text{row-span}\left(\mathbf{R}_{d,\boldsymbol{\mathcal{T}}(\mathbb{S})}\right)\cap\text{row-span}\left(\mathbf{R}_{d',\boldsymbol{\mathcal{T}}(\mathbb{S})}\right)$ is a sufficient condition for $\left\{\boldsymbol{c}^{\intercal}\boldsymbol{\theta}_d-\boldsymbol{c}^{\intercal}\boldsymbol{\theta}_{d'}\;\middle|\;\begin{array}{l}\left(\boldsymbol{\theta}_d,\boldsymbol{\pi}\right)\in\Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}\left(\kappa\right)\cap\tilde{\Pi}_{\mathbb{S}}\;,\\ \left(\boldsymbol{\theta}_{d'},\boldsymbol{\pi}\right)\in\Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d'}\left(\kappa\right)\cap\tilde{\Pi}_{\mathbb{S}}\end{array}\right\}$ and $\left\{\frac{\boldsymbol{c}^{\intercal}\boldsymbol{\theta}_d-\boldsymbol{c}^{\intercal}\boldsymbol{\theta}_{d'}}{\boldsymbol{c}^{\intercal}\boldsymbol{\pi}}\;\middle|\;\begin{array}{l}\left(\boldsymbol{\theta}_d,\boldsymbol{\pi}\right)\in\Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}\left(\kappa\right)\cap\tilde{\Pi}_{\mathbb{S}}\;,\\ \left(\boldsymbol{\theta}_{d'},\boldsymbol{\pi}\right)\in\Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d'}\left(\kappa\right)\cap\tilde{\Pi}_{\mathbb{S}}\end{array}\right\}$ to collapse to a point.

These conditions are not, in general, *necessary* for point-identification. In the absence of these conditions, point identification can still be obtained if some "knife-edge" conditions hold, as discussed in the following remark[9].

[9] See also Navjeevan, Pinto, and Santos (2023) for necessary and sufficient conditions for point identification in a broader class of models, including those considered here.

Remark 9 ("Knife-edge" Conditions for Point Identification)**.** Let $\mathbb{S}$ be a treatment selection model such that $\mathcal{Q}_{\mathbb{S}} \neq \emptyset$. If $\Pi_{\mathbb{S}} = \bigcap_{d\in\mathcal{D}} \Pi_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d} \cap \mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ is a singleton, say $\{\boldsymbol{p}\}$, then obviously $\boldsymbol{\pi} = \boldsymbol{p}$ is point-identified, but $\boldsymbol{\pi\xi}_{Y_d}$ may not be point-identified for any d. If $\boldsymbol{p}$ is a *vertex* of $\Pi_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}$, however, then $\boldsymbol{\pi\xi}_{Y_d}$ is point-identified! I sketch the argument below. A similar argument shows that $\boldsymbol{c}^{\mathsf{T}}\boldsymbol{\pi\xi}_{Y_d}$ is point-identified if $\bigcap_{d\in\mathcal{D}} \Pi_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d} \cap \mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ is a (non-empty) subset of the extreme face of the polytope $\Pi_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}$ with respect to the direction $\boldsymbol{c}$[10].

To see this, we can use the fact that any vertex of $\Pi_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}$ is the result of integrating along the corresponding vertex of $\mathcal{B}_d(\mathbf{pf}_d(y))$. Indeed, it is possible to show that the extreme face of the polytope $\Pi_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}$, with respect to a direction $\boldsymbol{c}$, is the following set

$$\left\{ \int_{\mathcal{Y}} \boldsymbol{g}(y)\, d\mu(y) \;\middle|\; \begin{array}{l} \boldsymbol{g} : \mathcal{Y} \to \mathbb{R}_{\geq 0}^{|\boldsymbol{\mathcal{T}}(\mathbb{S})|} \text{ is measurable,} \\ \boldsymbol{g}(y) \in \mathcal{B}_d^{\boldsymbol{c}}(\mathbf{pf}_d(y))\, \forall y \in \mathcal{Y} \end{array} \right\}, \tag{3}$$

where $\mathcal{B}_d^{\boldsymbol{c}}(\mathbf{pf}_d(y))$ is defined as the extreme face of the polytope $\mathcal{B}_d(\mathbf{pf}_d(y))$ with respect to direction $\boldsymbol{c}$[11]. Since $\boldsymbol{p}$ is a vertex of $\Pi_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}$, there exists a direction $\boldsymbol{c_p}$ such that $\operatorname{argmax}_{\boldsymbol{x}\in\Pi_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}} \{\boldsymbol{c_p^{\mathsf{T}}x}\} = \{\boldsymbol{p}\}$. Now, it is clear that (3) is a singleton if and only if $\mathcal{B}_d^{\boldsymbol{c_p}}(y)$ is a singleton, say $\left\{\boldsymbol{b}_d^{\boldsymbol{c_p}}(y)\right\}$, for every $y \in \mathcal{Y}$, which immediately gives us the result. Indeed, when this is the case, $\boldsymbol{\pi\xi}_{Y_d}(y) = \boldsymbol{b}_d^{\boldsymbol{c_p}}(y)$ for every $y \in \mathcal{Y}$. Of course, this also implies the point-identification of the corresponding mean counterfactuals.

3.1.2. *Joint Distribution of Counterfactual Outcomes.*

Theorem 6. *(Identified Set for Joint Outcome Distribution) For any treatment selection model $\mathbb{S}$, the identified set, under $\mathbb{S}$, for the conditional joint distributions of $(Y_d : d \in \mathcal{D})$ given $\boldsymbol{T}$ is as follows*

$$\begin{aligned} &\left\{ \left(Q_{(Y_d:d\in\mathcal{D})|\boldsymbol{T}=\boldsymbol{\tau}} : \boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}(\mathbb{S}) \right) \;\middle|\; Q \in \mathcal{Q}_{\mathbb{S}} \right\} \\ = &\left\{ \left(C_{\boldsymbol{\tau}}\left(\left(\mathfrak{F}\left[\boldsymbol{g}_{(d)(\boldsymbol{\tau})}\right] : d \in \mathcal{D} \right) \right) : \boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}(\mathbb{S}) \right) \;\middle|\; \begin{array}{l} \forall \boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}(\mathbb{S}), C_{\boldsymbol{\tau}} \textit{ is a } |\mathcal{D}|\textit{-dimensional copula}. \\ \forall d \in \mathcal{D}, \int_{\mathcal{Y}} \boldsymbol{g}_{(d)}\, d\mu(y) = \mathbf{1}_{|\boldsymbol{\mathcal{T}}(\mathbb{S})|}. \\ \exists \boldsymbol{\pi} \in \Pi_{\mathbb{S}} \text{ s.t. } \boldsymbol{\pi} \odot \boldsymbol{g}_{(d)} \in \boldsymbol{\Gamma}_d(\boldsymbol{\mathcal{T}}(\mathbb{S}))\ \forall d \in \mathcal{D}. \end{array} \right\}, \end{aligned}$$

where $\boldsymbol{g}_{(d)(\boldsymbol{\tau})}$ is the $\boldsymbol{\tau}$-th component of $\boldsymbol{g}_{(d)}$, $\mathfrak{F}\left(\boldsymbol{g}_{(d)(\boldsymbol{\tau})}\right)$ is the CDF of $\boldsymbol{g}_{(d)(\boldsymbol{\tau})}$, and $\odot$ denotes the coordinate-wise product of two vectors (Hadamard product).

Proof. Follows from the proof of Theorem 3. See Appendix A.3 for details. ☐

Theorem 6 shows that Condition **M** and together provide information about the *joint* distribution of potential outcomes, $(Y_d : d \in \mathcal{D})$, but do so only through restrictions on the conditional marginals, $Y_d \mid \boldsymbol{T} = \boldsymbol{\tau}$, and on the response-type probabilities. Intuitively, this follows from the fact that Condition **M** implies that treatment is randomly assigned conditional on response-type. This also suggests a trade-off: settings that allow for more response-types generate finer partitions of $\operatorname{supp}(\boldsymbol{V})$ and can, therefore, *potentially* provide more information on the dependence between outcomes; of course, if there is not enough exogenous variation, then any potential gains are offset by the increased degree of underidentification of the marginals and mixing weights (response-type probabilities) themselves.

[10]In other words, if $\bigcap_{d\in\mathcal{D}} \Pi_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d} \cap \mathcal{F}_{\boldsymbol{T}}(\mathbb{S}) \subseteq \operatorname{argmax}_{\boldsymbol{x}\in\Pi_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}} \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x}$.

[11]This follows from Proposition 1.2 in Billera and Sturmfels (1992), see the proof of Lemma 1 for details.

Remark 10. Theorem 6 also suggests that the framework here can be used to consider parameters of the form $E(Y_d \mid Y_{d'} \leq c)$. I do not pursue it further in this draft, but the basic idea is the same as the extrapolation extension considered in Remark 2: Theorem 6 can be viewed as saying that, conditional on $\boldsymbol{T}$, we do not have to worry about any "pointwise" restrictions on the distribution of $(Y_d, Y_{d'})$. So, for the purposes of the Y_d distribution, we can define types based on $(\boldsymbol{T}, \mathbb{1}\{Y_{d'} < c\})$, for which the identified set can be found using the methods above (it is a function of the $(Y_{d'}, \boldsymbol{T})$ distribution), and so, we can use the same technique used here to get the identified set for functionals of $(Y_d, \boldsymbol{T})$ to get the identified set for $(Y_d, (\boldsymbol{T}, \mathbb{1}\{Y_{d'} < c\}))$.

4. INFERENCE (BASIC OUTLINE, PRELIMINARY)

In this section I assume that the given treatment selection model of interest, $\mathbb{S}$, satisfies the following condition. In the remainder, I will continue to treat $\mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ as known.

Condition C. *$\mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ is a convex polytope.*

Any selection model that simply eliminates response-types (e.g., Ordered/Unordered/Vector/Partial Monotonicity) automatically satisfies this condition. More generally, Condition **C** is satisfied by any selection model $\mathbb{S}$ such that the (*a priori*) restrictions on the type distribution imposed by $\mathbb{S}$ can be characterized by a set of linear (in)equalities involving response-type probabilities[12]. The main utility of Condition **C** is tractable computation of the support function of $\Pi_{\mathbb{S}}$: $\Pi_{\mathbb{S}}$ is a convex polytope whenever $\mathbb{S}$ satisfies Condition **C**.

Remark 11. I conjecture that (for finite $\mathcal{Z}$ and $\mathcal{D}$), under Condition **M**, Assumption 2.1 in Lee and Salanié (2018) implies Condition **C**, but I have not been able to prove this (attempt in progress). If true, this would enable a broad class of threshold-crossing-rules-based models to be covered by the framework in this paper. I emphasize, however, that "translating" such models from the representation given in Lee and Salanié (2018) to a type restriction representation may be generally non-trivial.

4.1. Sharp Specification Test for Treatment Selection Models. [13]

To formalize our inference problem, let $\mathcal{P}$ be the set of all probability distributions on $\mathcal{Y} \times \mathcal{D} \times \mathcal{Z}$ satisfying the regularity conditions outlined in Section 2, and let

$$\mathcal{P}_{\mathbb{S}} := \{P \in \mathcal{P} : \mathcal{Q}_{\mathbb{S}}(P) \neq \emptyset\}.$$

Our hypothesis testing problem is

$$H_0 : P \in \mathcal{P}_{\mathbb{S}} \quad \text{against} \quad H_1 : P \in \mathcal{P} \setminus \mathcal{P}_{\mathbb{S}} \tag{4}$$

In order to use Theorem 3 to test if $P \in \mathcal{P}_{\mathbb{S}}$, we need to check

(i) For each $d \in \mathcal{D}$, for all $y \in \mathcal{Y}$, $\mathcal{B}_{\mathbf{R}_{d,\boldsymbol{T}(\mathbb{S})}}(\mathbf{pf}_d(y)) \neq \emptyset$

(ii) $\bigcap_{d \in \mathcal{D}} \Pi_{\boldsymbol{T}(\mathbb{S})|d}(P) \cap \mathcal{F}_{\boldsymbol{T}}(\mathbb{S}) \neq \emptyset$

[12] In this case, $\mathcal{F}_{\boldsymbol{T}}(\mathbb{S})$ is simply the intersection of the standard $\mathcal{D}^{\mathcal{Z}}$-simplex with the polytope defined by the non-negative solutions to such linear (in)equalities.

[13] The tests proposed in the next two subsections are preliminary and meant as a proof-of-concept. Several optimizations are possible, but I have not yet pursued them in this draft (but see Remark 12).

Of course, since $\Pi_{\mathcal{T}|d} = \emptyset$ whenever there is some $y \in \mathcal{Y}$ such that $\mathcal{B}_{\mathbf{R}_{d,\mathcal{T}}}(\mathbf{pf}_d(y)) = \emptyset$, (ii) implies (i). The distinction is important for implementation, however, because if (i) holds we can use Lemma 1 to construct a test of (ii).

Now, note that $\mathcal{B}_{\mathbf{R}_{d,\mathcal{T}(\mathbb{S})}}(\mathbf{pf}_d(y)) \neq \emptyset$ if and only if $\mathbf{pf}_d(y) \in \text{cone}(\mathbf{R}_d)$, where $\text{cone}(\mathbf{R}_d)$ is the cone generated by non-negative linear combinations of the column vectors of $\mathbf{R}_d$,

$$\text{cone}(\mathbf{R}_d) := \left\{ \boldsymbol{b} \in \mathbb{R}^{\mathcal{Z}} : \exists \boldsymbol{x} \in \mathbb{R}^{\mathcal{T}}_{\geq 0}, \mathbf{R}_d \boldsymbol{x} = \boldsymbol{b} \right\} .$$

In some cases, this step can be omitted because it reduces to testing $\mathbf{pf}_d \geq 0$ (which is automatically satisfied); for example, if the identity matrix is a sub-matrix of $\mathbf{R}_d$, then $\mathcal{B}_{\mathbf{R}_{d,\mathcal{T}(\mathbb{S})}}(\mathbf{pf}_d(y)) \neq \emptyset$ for all $y \in \mathcal{Y}$ is automatically satisfied. More generally, this is a polyhedral cone inclusion problem that can be represented as conditional moment inequality restrictions, as shown in the next result.

Lemma 3. *For any $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$ and $d \in \mathcal{D}$, there exist, for some $K_d \in \mathbb{N}$, known functions $\left\{ c^{(d;\mathcal{T})}_{z,j}(P_Z) \,\middle|\, z \in \mathcal{Z}, j \in \{1, \ldots, K_d\} \right\}$ such that*

$$\begin{aligned} & \mathcal{B}_{\mathbf{R}_{d,\mathcal{T}}}(\mathbf{pf}_d(y)) \neq \emptyset \ \forall y \in \mathcal{Y} \\ \iff & \max_{j \in \{1,\ldots,K_d\}} \sup_{y \in \mathcal{Y}} E_P \left[\sum_{z \in \mathcal{Z}} c^{(d;\mathcal{T})}_{z,j}(P_Z) \mathbb{1}\{D = d, Z = z\} \,\middle|\, Y = y \right] \leq 0 \, . \end{aligned}$$

Proof. See Appendix B.1. □

The proof of Lemma 3 generalizes the approach employed by Mourifié and Wan (2017) to test LATE assumptions in the binary treatment setting. Next, we have the following result for testing (ii), which implies that, conditional on $\mathcal{B}_{\mathbf{R}_{d,\mathcal{T}}}(\mathbf{pf}_d(y)) \neq \emptyset$ holding for all $y \in \mathcal{Y}$, (ii) can be tested by evaluating $\int_{\mathcal{Y}} h_{\mathbf{R}_1}(\boldsymbol{c}; \mathbf{pf}_1(y)) \, d\mu(y)$ at finitely many known values of $\boldsymbol{c}$, for each $d \in \mathcal{D}$. In practice, the matrices $\boldsymbol{M}_d$ can be found by row-reducing $\mathbf{R}_d$ and eliminating pivot variables.

Lemma 4. *For any treatment selection model $\mathbb{S}$ that satisfies Condition C, there exist known (i.e., known from model, does not depend on P) matrices $\boldsymbol{M}_d$, for $d \in \mathcal{D}$, known matrix $\boldsymbol{S}$, and known vector $\boldsymbol{s}$ such that*

$$\bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}(\mathbb{S})|d} \cap \mathcal{F}_{\mathcal{T}}(\mathbb{S}) \neq \emptyset \iff \exists \boldsymbol{x} \in [0,1]^{|\mathcal{T}(\mathbb{S})|} \text{ such that } \begin{bmatrix} \boldsymbol{M}_1 \\ \vdots \\ \boldsymbol{M}_{N_D} \\ \boldsymbol{S} \end{bmatrix} \boldsymbol{x} \leq \boldsymbol{h}_{\mathbb{S}}(P) \ ,$$

where

$$\boldsymbol{h}_{\mathbb{S}}(P) := \begin{bmatrix} \left(\int_{\mathcal{Y}} h_{\mathbf{R}_1}\left(\boldsymbol{M}_{1,(i)}; \mathbf{pf}_1(y)\right) \, d\mu(y) : i \in [|\text{rows}(\boldsymbol{M}_1)|] \right) \\ \vdots \\ \left(\int_{\mathcal{Y}} h_{\mathbf{R}_{N_D}}\left(\boldsymbol{M}_{N_D,(i)}; \mathbf{pf}_{N_D}(y)\right) \, d\mu(y) : i \in [|\text{rows}(\boldsymbol{M}_{N_D})|] \right) \\ \boldsymbol{s} \end{bmatrix} ,$$

and we use the convention that the support function of an empty set is constant at $-\infty$.

Proof. See Appendix B.1. □

Finally, I propose the following test statistic for specification testing treatment selection models.

Proposition 7. *For any treatment selection model $\mathbb{S}$ that satisfies Condition C, there exist known vectors $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_K$, for some $K \geq 1$, such that the hypothesis testing problem (4) is equivalent to testing*

$$H_0 : \vartheta_{\mathbb{S}} \leq 0 \quad \textit{against} \quad H_1 : \vartheta_{\mathbb{S}} > 0$$

where

$$\vartheta_{\mathbb{S}} := \max \left\{ \sup_{(y,d,j) \in \mathcal{Y} \times \mathcal{D} \times [K_d]} E_P \left[\sum_{z \in \mathcal{Z}} c_{z,j}^{(d;\boldsymbol{\mathcal{T}}(\mathbb{S}))} (P_Z) \mathbb{1}\{D = d, Z = z\} \,\middle|\, Y = y \right], \max_{1 \leq i \leq K} \boldsymbol{c}_i^{\mathsf{T}} \boldsymbol{h}_{\mathbb{S}}(P) \right\},$$

where the functions $c_{z,j}^{(d;\boldsymbol{\mathcal{T}}(\mathbb{S}))}$ are as given in Lemma 3, and $\boldsymbol{h}_{\mathbb{S}}$ is as given in Lemma 4.

Proof. See Appendix B.1. □

The main advantage of the specification in Proposition 7 is implementation, since they are of the form considered by the intersectional bounds framework of Chernozhukov, Lee, and Rosen (2013). To implement such a test in practice, we may need to restrict $\mathcal{P}$ further. For example, if Y is continuous, then we will generally at least require $\mathcal{P}$ to be such that $\mathbf{pf}_d$ has twice differentiable components and is bounded away from zero. Work to formalize these conditions is in progress.

4.2. **Inference on Mean Counterfactual Outcomes and Treatment Effects.** Fix some continuous $\kappa : \mathcal{Y} \to \mathbb{R}$ such that $\kappa(\mathcal{Y})$ is a convex compact set. Define

$$\boldsymbol{\theta}_d := (\mathrm{P}(\boldsymbol{T} = \boldsymbol{\tau}) E(\kappa(Y_d) \mid \boldsymbol{T} = \boldsymbol{\tau}) : \boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}(\mathbb{S})) ,$$

and let $\boldsymbol{\theta}_{d,Q} := (Q(\boldsymbol{T} = \boldsymbol{\tau}) E_Q(\kappa(Y_d) \mid \boldsymbol{T} = \boldsymbol{\tau}) : \boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}(\mathbb{S}))$.

In this section, I propose methodology for performing inference on parameters of the form

$$\frac{\sum_{d \in \mathcal{D}} \boldsymbol{c}_d^{\mathsf{T}} \boldsymbol{\theta}_d}{\boldsymbol{c}_\pi^{\mathsf{T}} \boldsymbol{\pi}} \ \text{ or } \ \sum_{d \in \mathcal{D}} \boldsymbol{c}_d^{\mathsf{T}} \boldsymbol{\theta}_d \ \text{ or } \ \boldsymbol{c}_\pi^{\mathsf{T}} \boldsymbol{\pi} ,$$

for some known $\boldsymbol{c}_d \in \mathbb{R}^{|\boldsymbol{\mathcal{T}}|}$, for $d \in \mathcal{D}$, and $\boldsymbol{c}_\pi \in \mathbb{R}^{|\boldsymbol{\mathcal{T}}|}$. Such parameters include (weighted/unweighted) treatment effects, outcome means, and response-type probabilities. For simplicity, I focus on the parameter

$$\beta := \frac{\boldsymbol{c}_d^{\mathsf{T}} \boldsymbol{\theta}_d - \boldsymbol{c}_{d'}^{\mathsf{T}} \boldsymbol{\theta}_{d'}}{\boldsymbol{c}_\pi^{\mathsf{T}} \boldsymbol{\pi}} ,$$

but the results extend analogously to the more general case. I assume that $\boldsymbol{c}_\pi^{\mathsf{T}} \boldsymbol{\pi} > 0$ for all $\boldsymbol{\pi} \in \Pi_{\mathbb{S}}$.

A simple corollary of Theorem 4 is that

$$\Theta_{\mathbb{S}}(\beta) := \left\{ \frac{\boldsymbol{c}_d^{\mathsf{T}} \boldsymbol{\theta}_{d,Q} - \boldsymbol{c}_{d'}^{\mathsf{T}} \boldsymbol{\theta}_{d',Q}}{\boldsymbol{c}_\pi^{\mathsf{T}} \boldsymbol{\pi}_Q} \,\middle|\, Q \in \mathcal{Q}_{\mathbb{S}} \right\} = \left\{ \frac{\boldsymbol{c}_d^{\mathsf{T}} \boldsymbol{\theta}_d - \boldsymbol{c}_{d'}^{\mathsf{T}} \boldsymbol{\theta}_{d'}}{\boldsymbol{c}_\pi^{\mathsf{T}} \boldsymbol{\pi}} \,\middle|\, \begin{array}{l} (\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}} , \\ (\boldsymbol{\theta}_{d'}, \boldsymbol{\pi}) \in \Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d'}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}} \end{array} \right\} .$$

Let $\mathcal{P}_{\mathbb{S}} = \{P \in \mathcal{P} : \mathcal{Q}_{\mathbb{S}}(P) \neq \emptyset\}$, as in the previous subsection, and define $\mathcal{P}_{\mathbb{S}}(\beta;\gamma)$, for each $\gamma \in \mathbb{R}$, as

$$\mathcal{P}_{\mathbb{S}}(\beta;\gamma) := \{P \in \mathcal{P}_{\mathbb{S}} : \gamma \in \Theta_{\mathbb{S}}(\beta)\} \ .$$

For a given value of $\gamma \in \mathbb{R}$, we are interested in the following hypothesis testing problem

$$H_0 : P \in \mathcal{P}_{\mathbb{S}}(\beta;\gamma) \qquad \text{against} \qquad H_1 : P \in \mathcal{P}_{\mathbb{S}} \setminus \mathcal{P}_{\mathbb{S}}(\beta;\gamma) \tag{5}$$

To proceed, first note that

$$\Theta_{\mathbb{S}}(\beta) = \left\{ b \in \mathbb{R} \,\middle|\, \begin{array}{l} \exists\,(\boldsymbol{\theta}_d, \boldsymbol{\pi}_d, \boldsymbol{\theta}_{d'}, \boldsymbol{\pi}_{d'}) \in \Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}(\kappa) \times \Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d'}(\kappa) \ \text{s.t.}\ , \\ \boldsymbol{\pi}_d = \boldsymbol{\pi}_{d'} \in \Pi_{\mathbb{S}}, \\ \boldsymbol{c}_d^{\intercal}\boldsymbol{\theta}_d - \boldsymbol{c}_{d'}^{\intercal}\boldsymbol{\theta}_{d'} = b\boldsymbol{c}_\pi^{\intercal}\boldsymbol{\pi}_d \end{array} \right\} .$$

Next, motivated by this definition, define

$$\Lambda_{\mathbb{S}}(\beta;\gamma) := \left\{ (\boldsymbol{\theta}_d, \boldsymbol{\pi}_d, \boldsymbol{\theta}_{d'}, \boldsymbol{\pi}_{d'}) \,\middle|\, \begin{array}{l} \boldsymbol{\theta}_d, \boldsymbol{\theta}_{d'} \in \kappa(\mathcal{Y})^{|\boldsymbol{\mathcal{T}}|},\ \boldsymbol{\pi}_d, \boldsymbol{\pi}_{d'} \in \Pi_{\mathbb{S}} \\ \boldsymbol{\pi}_d = \boldsymbol{\pi}_{d'},\ \boldsymbol{c}_d^{\intercal}\boldsymbol{\theta}_d - \boldsymbol{c}_{d'}^{\intercal}\boldsymbol{\theta}_{d'} = \gamma\boldsymbol{c}_\pi^{\intercal}\boldsymbol{\pi}_d \end{array} \right\} .$$

Note that when $\mathbb{S}$ satisfies Condition C, $\Lambda_{\mathbb{S}}(\beta;\gamma)$ is a compact convex polytope. Denote by $h_{\Lambda_{\mathbb{S}}(\beta;\gamma)}$ the support function of $\Lambda_{\mathbb{S}}(\beta;\gamma)$. Now, we have the following result.

Proposition 8. *For any treatment selection model $\mathbb{S}$ that satisfies Condition C, the hypothesis testing problem* (5) *is equivalent to*

$$H_0 : \sup_{\boldsymbol{v} \in S^{4|\boldsymbol{\mathcal{T}}(\mathbb{S})|}} \vartheta_{(\beta;\gamma)}(\boldsymbol{v}) \leq 0 \ \textit{ against } \ H_1 : \sup_{\boldsymbol{v} \in S^{4|\boldsymbol{\mathcal{T}}(\mathbb{S})|-1}} \vartheta_{(\beta;\gamma)}(\boldsymbol{v}) > 0 \ ,$$

where $S^{4|\boldsymbol{\mathcal{T}}(\mathbb{S})|-1}$ is the unit sphere in $\mathbb{R}^{4|\boldsymbol{\mathcal{T}}(\mathbb{S})|}$, and

$$\begin{aligned} \vartheta_{(\beta;\gamma)}(\boldsymbol{v}) := &-\int_{\mathcal{Y}} h_{\mathbf{R}_d}\big(\kappa(y)\boldsymbol{v}_1 + \boldsymbol{v}_2\ ;\ \mathbf{pf}_d(y)\big)\, d\mu(y) \\ &- \int_{\mathcal{Y}} h_{\mathbf{R}_{d'}}\big(\kappa(y)\boldsymbol{v}_3 + \boldsymbol{v}_4\ ;\ \mathbf{pf}_{d'}(y)\big)\, d\mu(y) \\ &- h_{\Lambda_{\mathbb{S}}(\beta;\gamma)}(-\boldsymbol{v}) \ , \end{aligned}$$

for all $\boldsymbol{v} = (\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3, \boldsymbol{v}_4) \in S^{4|\boldsymbol{\mathcal{T}}(\mathbb{S})|}$ with $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3, \boldsymbol{v}_4 \in \mathbb{R}^{|\boldsymbol{\mathcal{T}}(\mathbb{S})|}$.

Proof. See Appendix B.2. □

The test proposed in Proposition 8 can be inverted to produce confidence regions for β. Note that it is once again of the intersectional bounds form considered by Chernozhukov et al. (2013) (CLR) and, under some regularity conditions, this test can be implemented using their framework. Note also that if we replace the test statistic in Proposition 8 with the maximum of the test statistics in Propositions 7 and 8, then the joint test will always fail to reject whenever the specification test fails to reject, and so, will return an empty confidence region whenever we can reject $P \in \mathcal{P}_{\mathbb{S}}$.

Remark 12. Implementing the test proposed in Proposition 8 using CLR's framework may be very computationally demanding when there are many response-types because their approach will typically involve optimization over a grid on $S^{4|\boldsymbol{\mathcal{T}}(\mathbb{S})|-1}$. But $\vartheta_{(\beta;\gamma)}(\boldsymbol{v})$ is a concave function, and so, computation of the test statistic $\sup_{\boldsymbol{v} \in S^{4|\boldsymbol{\mathcal{T}}(\mathbb{S})|}} \vartheta_{(\beta;\gamma)}(\boldsymbol{v})$ is a convex programming problem. In our setting, however, implementation using CLR's methodology generally cannot take advantage of this fact. This is because their approach rejects H_0 for large values of the following *precision-corrected* estimate of the test statistic,

$$\hat{\vartheta}_0(p) := \sup_{\boldsymbol{v}\in S^{4|\boldsymbol{\mathcal{T}}(\mathbb{S})|}} \left(\hat{\vartheta}_{(\beta;\gamma)}(\boldsymbol{v}) - k(p)\, s(\boldsymbol{v})\right) ,$$

where $\hat{\vartheta}_{(\beta;\gamma)}(\boldsymbol{v})$ is an estimator for $\vartheta_{(\beta;\gamma)}(\boldsymbol{v})$, $k(p)$ is a critical value, and $s(\boldsymbol{v})$ is a (pointwise) standard error for $\hat{\vartheta}_{(\beta;\gamma)}(\boldsymbol{v})$. If $\hat{\vartheta}_{(\beta;\gamma)}(\boldsymbol{v})$ is a plug-in estimator for $\vartheta_{(\beta;\gamma)}(\boldsymbol{v})$ based on, e.g., a kernel density estimator, then $\hat{\vartheta}_{(\beta;\gamma)}(\boldsymbol{v})$ remains a concave function, but the addition of the adjustment term $-k(p)\, s(\boldsymbol{v})$ generally renders $\hat{\vartheta}_{(\beta;\gamma)}(\boldsymbol{v}) - k(p)\, s(\boldsymbol{v})$ non-concave.

The precision-correction in CLR is due to the fact that, in the problems that motivate their work, the analog estimator $\sup_{\boldsymbol{v}\in S^{4|\boldsymbol{\mathcal{T}}(\mathbb{S})|}} \hat{\vartheta}_{(\beta;\gamma)}(\boldsymbol{v})$ can be severely biased upward in finite samples. Informally, this is because in settings like testing conditional moment inequalities (e.g.), the bound function $\vartheta_{(\beta;\gamma)}(\boldsymbol{v})$ has a "local character" (this is also true, for example, for the test statistic proposed in Proposition 7). On the other hand, this may not be a concern in our setting since the estimation of each of the three components of $\vartheta_{(\beta;\gamma)}(\boldsymbol{v})$ is perhaps best understood as the estimation of a finite dimensional polytope, which we should not expect to have such problems. Indeed, it is possible to show that the estimator, $\hat{h}_{\mathbf{R}_d}$, for $h_{\mathbf{R}_d}$ constructed by plugging in a suitably undersmoothed (and sufficiently high order) kernel density estimator, $\hat{\mathbf{pf}}_d$, for $\mathbf{pf}_d$, consistently (and uniformly) estimates the function $h_{\mathbf{R}_d}$ and $\sqrt{n}\left(\hat{h}_{\mathbf{R}_d} - h_{\mathbf{R}_d}\right)$, as a stochastic process, converges in law to a (non-degenerate) asymptotic distribution. This suggests that strong approximation methods are not necessary to implement a test based on Proposition 8. I am currently working to formalize an implementation based on the Delta method for directionally differentiable functionals, as provided in Fang and Santos (2019), and the numerical derivative approach developed in Hong and Li (2018), using the kernel CDF[14] as the underlying basis for the functional central limit theorem (as in, e.g., Aït-Sahalia, 1993). As suggested by Hong and Li (2018), this approach may also lead to uniformly valid inference. Formal results are pending and I do not consider this approach any further in this draft.

5. IMPLEMENTATION (PRELIMINARY)

To implement the identification results obtained above in practice, we need a computationally tractable approach to compute the support functions given by Lemmas 1 and 2:

$$h_{\Pi_{\boldsymbol{\mathcal{T}}|d}}(\boldsymbol{c}) = \int_{\mathcal{Y}} h_{\mathbf{R}_d}(\boldsymbol{c};\ \mathbf{pf}_d(y))\ d\mu(y)\ , \tag{6}$$

$$\text{and } h_{\Theta_{\boldsymbol{\mathcal{T}}|d}(\kappa)}(\boldsymbol{c}_1, \boldsymbol{c}_2) = \int_{\mathcal{Y}} h_{\mathbf{R}_d}\big(\kappa(y)\, \boldsymbol{c}_1 + \boldsymbol{c}_2\ ;\ \mathbf{pf}_d(y)\big)\ d\mu(y)\ . \tag{7}$$

The expressions above involve the function $(\boldsymbol{c}, y) \mapsto h_{\mathbf{R}_d}(\boldsymbol{c};\ \mathbf{pf}_d(y))$, which is essentially a nuisance parameter in our setting. For fixed $\boldsymbol{c} \in \mathbb{R}^{|\boldsymbol{\mathcal{T}}|}$ and $y \in \mathcal{Y}$, $h_{\mathbf{R}_d}(\boldsymbol{c};\ \mathbf{pf}_d(y))$ is a "small" linear programming problem. Keeping $\boldsymbol{c}$ fixed, the function $y \mapsto h_{\mathbf{R}_d}(\boldsymbol{c};\ \mathbf{pf}_d(y))$ is the solution to a *parametric* linear programming problem (parametric in the right hand side vector). When there are only a handful of distinct instrument levels, the problem can be solved symbolically (either "manually" using, e.g., Fourier-Motzkin elimination, or using a symbolic solver like Mathematica or Maple). Such analytic solutions are useful for

[14]Defined as $y \mapsto \int_{[\min \mathcal{Y}, y]} \hat{\mathbf{pf}}_d(u)\ d\mu(u)$.

developing intuition on the source of identification power, for performing inference, and for implementing sophisticated forms of covariate adjustment.

As in many settings, the analytic approach becomes increasingly complicated as the parameter space grows richer[15]. Indeed, the complexity of the analytic solution (and the difficulty of deriving it) generally increases exponentially with the cardinality of $\mathcal{Z}$, and it can be computationally prohibitive to solve symbolically when there are many instruments/response-types[16]. In such cases, $h_{\mathbf{R}_d}$ can still be easily approximated using a primal-dual approach (with deterministic bounds on the approximation error); in this section, I provide a general algorithm to implement such an approach for computing $h_{\mathbf{R}_d}$. Importantly, the proposed algorithm can provide an *exact solution after finitely many steps*[17]. When there are many instruments, however, the number of steps may be too high for practical purposes, and we may wish to rely on approximations of $h_{\mathbf{R}_d}$.

5.1. **Computation of $h_{\mathbf{R}_d}$.**

Notation and Preliminaries: Basic Solutions to Linear Programs. For some $\boldsymbol{b} \in \mathbb{R}^m$, $\boldsymbol{c} \in \mathbb{R}^n$, and $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, with $\operatorname{rank}(\boldsymbol{A}) = m$, consider the following linear program (LP):

$$\max\ \boldsymbol{c}^\intercal \boldsymbol{x} \quad \text{subject to} \quad \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \geq \boldsymbol{0}$$

For each subset $B \subseteq \{1, \ldots, n\}$ of columns of $\boldsymbol{A}$, let $\boldsymbol{A}_B$ be the sub-matrix formed by the columns in B. If $\boldsymbol{A}_B^{-1}$ is invertible, we will say that B is a basis. For any basis $B = \{j_1, \ldots, j_m\}$, with $j_1 < j_2 < \cdots < j_m$, such that $\boldsymbol{A}_B^{-1}\boldsymbol{b} \geq 0$, we can construct a feasible point $\boldsymbol{x}$, as follows: set $\boldsymbol{x}_i = 0$ for all $i \notin B$, and set each $\boldsymbol{x}_{j_k}$ to the k-the value of $\boldsymbol{A}_B^{-1}\boldsymbol{b}$. Such a feasible point is called a *basic feasible solution*. Suppose the LP has at least one solution, and that $\boldsymbol{c}^\intercal \boldsymbol{x}$ is bounded over the feasible space. Then, there always exists a basis $B^*(\boldsymbol{c}; \boldsymbol{b})$ such that the basic feasible solution associated with the basis $B^*(\boldsymbol{c}; \boldsymbol{b})$ is an *optimal* solution to the above LP. This is a well-known result in linear programming (see, e.g., Theorem 4.2.3 in Matousek and Gärtner, 2007). This result also continues to hold if the LP above is a minimization problem instead of a maximization problem.

The LP above (called the *primal* LP) has an associated *dual* LP:

$$\min\ \boldsymbol{b}^\intercal \boldsymbol{y} \quad \text{subject to} \quad \boldsymbol{A}^\intercal \boldsymbol{y} - \boldsymbol{s} = \boldsymbol{c},\ \boldsymbol{s} \geq \boldsymbol{0}\ ,$$

[15] See, for example, the brief discussion in Torgovitsky (2019a)

[16] Analytic solutions may still remain feasible as the number of instruments increase if we also impose increasingly severe restrictions on response-types, so that the "degree of underidentification" remains controlled.

[17] Both the symbolic and computational approaches are tackling a finite problem: polytopes of the form $K_{\boldsymbol{b}} := \{\boldsymbol{x} \geq 0 : \mathbf{R}_d \boldsymbol{x} = \boldsymbol{b}\}$ will take on one of finitely many "shapes" as the right hand side vector $\boldsymbol{b}$ is allowed to vary. More precisely, every polytope $K_{\boldsymbol{b}}$ generates a (finite) decomposition (called the normal fan of $K_{\boldsymbol{b}}$) of the space of linear functionals on $K_{\boldsymbol{b}}$ into functionals that are maximized on the same face of $K_{\boldsymbol{b}}$. If $K_{\boldsymbol{b}}$ and $K_{\boldsymbol{b}'}$ generate the same decomposition, they are said to be *normally equivalent* (they have the same "shape"); when this is the case, we can view maximizing $\boldsymbol{c}^\intercal \boldsymbol{x}$ over $K_{\boldsymbol{b}}$ and over $K_{\boldsymbol{b}'}$ as the "same" problem (in a formal sense). The "shape" of $K_{\boldsymbol{b}}$ corresponds to which inequalities are non-redundant in the halfspace representation of $K_{\boldsymbol{b}}$, and so, the number of possible shapes generally grows with $\mathcal{Z}$. For more details, I refer the interested reader to Section 1.2 in De Loera, Rambau, and Santos (2010), and references therein.

Since each decomposition is finite and since, for any fixed matrix $\mathbf{R}_d$, the space of right hand side vectors only generate finitely many "shapes", the problem of computing $h_{\mathbf{R}_d}(\cdot\,;\,\cdot)$ is finite (at least when $\mathbf{pf}_d$ is continuous and $\mathcal{Y}$ compact). Although both approaches deal with finite problems, the symbolic solution amounts to solving the problem for all possible observed data distributions, whereas the computational approach only needs to solve it for the actually observed data distribution (which may generate far fewer "shapes" than the maximum possible).

which can equivalently be represented in the same form as the original LP by writing:

$$\min \begin{bmatrix} \boldsymbol{b} \\ -\boldsymbol{b} \\ \boldsymbol{0} \end{bmatrix}^{\mathsf{T}} \begin{bmatrix} \boldsymbol{y}_1 \\ \boldsymbol{y}_2 \\ \boldsymbol{s} \end{bmatrix} \quad \text{subject to} \quad \begin{bmatrix} \boldsymbol{A}^{\mathsf{T}} & -\boldsymbol{A}^{\mathsf{T}} & -\boldsymbol{I}_n \end{bmatrix} \begin{bmatrix} \boldsymbol{y}_1 \\ \boldsymbol{y}_2 \\ \boldsymbol{s} \end{bmatrix} = \boldsymbol{c}, \ \begin{bmatrix} \boldsymbol{y}_1 \\ \boldsymbol{y}_2 \\ \boldsymbol{s} \end{bmatrix} \geq \boldsymbol{0} \,,$$

and so, the notion of optimal basis also makes sense for the dual LP. The weak duality theorem, a well-known result in linear programming, states that for every feasible solution, $\boldsymbol{x}$, to the primal LP, and every feasible solution, $\boldsymbol{y}$, to the dual LP, we have that $\boldsymbol{c}^{\mathsf{T}}\boldsymbol{x} \leq \boldsymbol{b}^{\mathsf{T}}\boldsymbol{y}$. Further, the strong duality theorem states that at optimal solutions, $\boldsymbol{x}^*$ and $\boldsymbol{y}^*$, to the primal and dual (respectively), we have that $\boldsymbol{c}^{\mathsf{T}}\boldsymbol{x}^* = \boldsymbol{b}^{\mathsf{T}}\boldsymbol{y}^*$.

In our case, we are interested in LPs of the form

$$\max \ \varkappa(y)^{\mathsf{T}} \boldsymbol{x} \quad \text{subject to} \quad \mathbf{R}_d \boldsymbol{x} = \mathbf{pf}_d(y) \,, \ \boldsymbol{x} \geq \boldsymbol{0} \,,$$

where $\varkappa : \mathcal{Y} \to \mathbb{R}^{|\boldsymbol{\mathcal{T}}|}$ is a known function, generally of the form $\varkappa(y) = \boldsymbol{c}$ or $\varkappa = \kappa(y)\,\boldsymbol{c}_1 + \boldsymbol{c}_2$. Denote by $B^{\text{prim}}_{\varkappa}(y)$ the optimal basis associated with this problem, and denote by $B^{\text{dual}}_{\varkappa}(y)$ the optimal basis associated with the dual of this problem; let $\mathbf{S}^{\text{prim}}_{d,B^{\text{prim}}_{\varkappa}(y)}$ and $\mathbf{S}^{\text{dual}}_{d,B^{\text{dual}}_{\varkappa}(y)}$ be the matrices such that the primal and dual solutions are given by $\mathbf{S}^{\text{prim}}_{d,B^{\text{prim}}_{\varkappa}(y)}\mathbf{pf}_d(y)$ and $\mathbf{S}^{\text{dual}}_{d,B^{\text{dual}}_{\varkappa}(y)}\varkappa(y)$, respectively. The primal and dual optimal bases can be found simultaneously via linear programming[18]; for any fixed y, the corresponding LP is very small in the settings that motivate this paper.

5.1.1. *Generic Computation Algorithm.* For simplicity, I focus on the case where Y is a scalar and $\mathcal{Y} \subseteq \mathbb{R}$ is a compact interval.

Remark 13. The algorithm proposed here is preliminary and meant as a proof-of-concept. Many optimizations are possible (for example, using a parametric LP approach), but I have not yet pursued them in this draft. The algorithm also abstracts away from how the actual linear programming step is performed, but certain kinds of "pre-processing" (e.g., eliminating redundant response-types etc.), can often significantly reduce the size of the problem.

Algorithm 1.

Input: Model and Data: binary matrix $\mathbf{R}_d$ and continuous function $\mathbf{pf}_d : \mathcal{Y} \to \text{cone}(\mathbf{R}_d)$.
Parameters: continuous function $\varkappa : \mathcal{Y} \to \mathbb{R}^{|\boldsymbol{\mathcal{T}}|}$, real number $\delta \geq 0$, and finite set of grid points $\mathcal{G} \subseteq \mathcal{Y}$ with $\{\min \mathcal{Y}, \max \mathcal{Y}\} \in \mathcal{G}$.

Output: $\left(\boldsymbol{h}^{\text{prim}}, \boldsymbol{h}^{\text{dual}}\right)$, two measurable functions in $\{\mathcal{Y} \to \mathbb{R}\}$.

Step 0. Set $\mathcal{G}_1 := \mathcal{G}$.

$\vdots$

Step j. For each $y \in \mathcal{G}_j$, find $B^{\text{prim}}_{\varkappa}(y)$ and $B^{\text{dual}}_{\varkappa}(y)$[19].

[18] Throughout this section, I assume that the LPs of interest are bounded, and therefore have solutions; while this is not necessarily true (the LP will be unbounded iff the objective function assigns a positive coefficient to a response-type that never takes treatment d), the degenerate situations that involve such problems are easy to handle separately.

[19] This is the linear programming step, and can be performed with any linear program solver that outputs the optimal bases (in addition to the optimal solution), e.g., Gurobi.

For each $y \in \mathcal{Y}$ define $\lfloor y \rfloor, \lceil y \rceil \in \mathcal{G}_j$ as the tightest values such that $\lfloor y \rfloor \leq y \leq \lceil y \rceil$, and define $\boldsymbol{s}_y := \max_{\tilde{y} \in \{\lfloor y \rfloor, \lceil y \rceil\}} \left\{ \min \left(\mathbf{S}^{\text{prim}}_{d, B^{\text{prim}}_{\varkappa}(\tilde{y})} \mathbf{pf}_d\,(y) \right) \right\}$.

(Feasibility Check) If $\inf_{y \in \mathcal{Y}} \boldsymbol{s}_y < 0$:
There exists $y_0 \leq \ldots, \leq y_k \in \mathcal{Y}$, for some $k \geq 1$, such that

$$\{y \in \mathcal{Y} \mid \boldsymbol{s}_y < 0\} = \bigcup_{n=1}^{k} (y_{n-1}, y_n) \ .$$

Set $\mathcal{G}_{j+1} := \mathcal{G}_j \cup \left\{ \frac{y_n - y_{n-1}}{2} : n \in [k] \right\}$, and immediately go to the next step.
Else, if $\inf_{y \in \mathcal{Y}} \boldsymbol{s}_y \geq 0$:
Order elements of $\mathcal{G}_j$ as $\inf \mathcal{Y} =: y_0 \leq \cdots \leq y_N := \sup \mathcal{Y}$ and compute

$$\boldsymbol{h}^{\text{prim}}_j (y) := \sum_{n=1}^{N} \mathbb{1} \{ y \in [y_{n-1}, y_n] \} \max_{\tilde{y} \in \{\lfloor y \rfloor, \lceil y \rceil\} : \boldsymbol{s}_{\tilde{y}} \geq 0} \left\{ \varkappa (y)^\intercal \, \mathbf{S}^{\text{prim}}_{d, B^{\text{prim}}_{\varkappa}(\tilde{y})} \mathbf{pf}_d\,(y) \right\} \ ,$$

$$\boldsymbol{h}^{\text{dual}}_j (y) := \sum_{n=1}^{N} \mathbb{1} \{ y \in [y_{n-1}, y_n] \} \min_{\tilde{y} \in \{\lfloor y \rfloor, \lceil y \rceil\} : \boldsymbol{s}_{\tilde{y}} \geq 0} \left\{ \mathbf{pf}_d\,(y)^\intercal \, \mathbf{S}^{\text{dual}}_{d, B^{\text{dual}}_{\varkappa}(\tilde{y})} \varkappa (y) \right\} \ .$$

(Optimality Check) If $\sup_{y \in \mathcal{Y}} \left(\boldsymbol{h}^{\text{dual}}_j (y) - \boldsymbol{h}^{\text{prim}}_j (y) \right) > \delta$:
There exists $y_0 \leq \ldots, \leq y_k$, for some $k \geq 1$, such that

$$\left\{ y \in \mathcal{Y} \,\middle|\, \left(\boldsymbol{h}^{\text{dual}}_j (y) - \boldsymbol{h}^{\text{prim}}_j (y) \right) > \delta \right\} = \bigcup_{n=1}^{k} (y_{n-1}, y_n) \ .$$

Set $\mathcal{G}_{j+1} := \mathcal{G}_j \cup \left\{ \frac{y_n - y_{n-1}}{2} : n \in [k] \right\}$, and go to the next step.

Else, if $\sup_{y \in \mathcal{Y}} \left(\boldsymbol{h}^{\text{dual}}_j (y) - \boldsymbol{h}^{\text{prim}}_j (y) \right) \leq \delta$: Output $\left(\boldsymbol{h}^{\text{prim}}_j, \boldsymbol{h}^{\text{dual}}_j \right)$ and stop.

$\vdots$

Now, we have the following result.

Proposition 9. *Assume that $\mathcal{Y} \subseteq \mathbb{R}$ is a compact interval, that μ is the Lebesgue measure, and that $\mathbf{pf}_d$ is continuous. For any treatment selection model $\mathbb{S}$ such that $\mathcal{B}_d\,(\mathbf{pf}_d\,(y)) \neq \emptyset$ for all $y \in \mathcal{Y}$, and any continuous function $\varkappa : \mathcal{Y} \to \mathbb{R}^{|\mathcal{T}|}$, real number $\delta \geq 0$, and finite set of grid points $\mathcal{G} \subseteq \mathcal{Y}$ such that $\{\min \mathcal{Y}, \max \mathcal{Y}\} \in \mathcal{G}$, the following statements hold for Algorithm 1:*

(i) *Algorithm 1 will terminate in finitely many steps, and output two measurable functions $\left(\boldsymbol{h}^{prim}, \boldsymbol{h}^{dual} \right)$ in $\{\mathcal{Y} \to \mathbb{R}\}$ such that*

$$\sup_{y \in \mathcal{Y}} \left(\boldsymbol{h}^{dual} (y) - \boldsymbol{h}^{prim} (y) \right) \leq \delta$$

(ii) *Algorithm 1's output $\left(\boldsymbol{h}^{prim}, \boldsymbol{h}^{dual} \right)$ will always be such that, for all $y \in \mathcal{Y}$*

$$\boldsymbol{h}^{prim} (y) \leq h_{\mathbf{R}_d} \left(\varkappa (y) ; \ \mathbf{pf}_d\,(y) \right) \leq \boldsymbol{h}^{dual} (y)$$

Further, for any such selection model, for every $(\boldsymbol{c}_1, \boldsymbol{c}_2) \in \mathbb{R}^{|\mathcal{T}|} \times \mathbb{R}^{|\mathcal{T}|}$ *and every continuous* $\kappa : \mathcal{Y} \to \mathbb{R}$*, there exists a finite set* $\mathcal{G}^* \subseteq \mathcal{Y}$ *such that with input parameters* $\varkappa(y) = \kappa(y)\, \boldsymbol{c}_1 + \boldsymbol{c}_2,\ \delta = 0$*, and* $\mathcal{G} = \mathcal{G}^*$*, Algorithm 1 stops after Step 1.*

Proof. See Appendix C.1. □

Remark 14. In practice, we do not know $\mathbf{pf}_d$, and so, we will need to implement Algorithm 1 using an estimator, $\hat{\mathbf{pf}}_d$, for $\mathbf{pf}_d$ (e.g. kernel density estimator). It may sometimes be the case that $\hat{\mathbf{pf}}_d \notin \operatorname{cone}(\mathbf{R}_d)$ even though $\mathbf{pf}_d \in \operatorname{cone}(\mathbf{R}_d)$, which may cause problems with Algorithm 1. In such cases, Algorithm 1 can be implemented after projecting $\hat{\mathbf{pf}}_d$ onto $\operatorname{cone}(\mathbf{R}_d)$. Note that although the specification test proposed in Proposition 7 requires the computation of quantities like (6), it is not required for the test proposed in Lemma 3 for testing $\mathcal{B}_d(\mathbf{pf}_d(y)) \neq \emptyset\ \forall y \in \mathcal{Y}$ (or, equivalently, $\mathbf{pf}_d \in \operatorname{cone}(\mathbf{R}_d)$), which is all that is necessary for the efficacy of Algorithm 1.

6. CONCLUSION

This paper presents new specification testing and partial identification results for instrumental variable models with multiple levels (or types) of treatment and discrete-valued, randomly assigned, instruments. The proposed framework allows for the partial identification of a broad class of causal parameters under many different kinds of assumptions about the treatment selection mechanism. Other than random assignment of the instrument, I do not require any additional restrictions on treatment selection, but allow the analyst to flexibly model selection into treatment. Using a novel geometric characterization of IV mixture models, this paper provides a characterization of the empirical content and testable implications of such models, and a a new characterization of the identified set for counterfactual choices, outcome means, and treatment effects under such models. I propose methods for (i) (sharply) testing treatment selection model specifications and (ii) performing inference on mean causal effects and counterfactual outcomes under such assumptions. Finally, I propose a computationally feasible algorithm for implementing the proposed method without relying on any parametric or semiparametric assumptions.

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APPENDIX A: PROOFS OF MAIN IDENTIFICATION RESULTS

A.1. Proof of Theorem 1

Theorem 1. *For any treatment selection model* $\mathbb{S}$*,*

$$\mathcal{Q}_{\mathbb{S}} = \{Q \in \mathcal{Q} \mid \boldsymbol{\pi}_Q \in \mathcal{F}_{\boldsymbol{T}}(\mathbb{S})\} \ .$$

Proof. The "$\subseteq$" direction is clear, and so, it suffices to show "$\supseteq$".

Fix $Q \in \{Q \in \mathcal{Q} : Q_{\boldsymbol{T}} \in \mathcal{F}_{\boldsymbol{T}}(\mathbb{G}, \boldsymbol{\mathcal{V}}, \mathcal{F}_{\boldsymbol{V}})\}$ and define $\boldsymbol{\mathcal{T}}_Q := \left\{\boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}} : Q(\boldsymbol{T} = \boldsymbol{\tau}) > 0\right\}$. We have that there exists $\tilde{Q}$ such that $\tilde{Q}_{\boldsymbol{T}} = Q_{\boldsymbol{T}}$ and $\tilde{Q}$ satisfies $(\mathbb{G}, \boldsymbol{\mathcal{V}}, \mathcal{F}_{\boldsymbol{V}})$. So, there exists a measurable function $g_D \in \mathbb{G}$ and random variable $\boldsymbol{V}$ with $\operatorname{supp}(\boldsymbol{V}) = \boldsymbol{\mathcal{V}}$ and $\operatorname{Law}(\boldsymbol{V}) \in \mathcal{F}_{\boldsymbol{V}}$ such that $\operatorname{Law}((g_D(z, \boldsymbol{V}) : z \in \mathcal{Z})) = \tilde{Q}_{\boldsymbol{T}} = Q_{\boldsymbol{T}}$. Let $Z \sim Q_Z$ with $Z \perp \boldsymbol{V}$ mutually independent.

Now it suffices to show that there exists measurable function g_Y and random variable $\boldsymbol{\varepsilon}_Y$ such that $Z, \boldsymbol{V}, \boldsymbol{\varepsilon}_Y$ are mutually independent, and, for each $\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}_Q$,

$$\operatorname{Law}((g_Y(d, \boldsymbol{V}, \boldsymbol{\varepsilon}_Y) : d \in \mathcal{D}) \mid \boldsymbol{T} = \boldsymbol{\tau}) = Q_{(Y_d : d \in \mathcal{D}) \mid \boldsymbol{T} = \boldsymbol{\tau}} \ .$$

For each $\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}_Q$, define $\boldsymbol{\varepsilon}_Y(\boldsymbol{\tau})$ as a $|\mathcal{D}|$-dimensional random vector with distribution $Q_{(Y_d : d \in \mathcal{D}) \mid \boldsymbol{T} = \boldsymbol{\tau}}$, with $\boldsymbol{\varepsilon}_Y := (\boldsymbol{\varepsilon}_Y(\boldsymbol{\tau})^\intercal : \boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}_Q)^\intercal$ independent from $(Z, \boldsymbol{V})$. Next, define $g_Y : \mathcal{D} \times \operatorname{supp}(\boldsymbol{V}) \times \operatorname{supp}(\boldsymbol{\varepsilon}_Y)$

$$g_Y(d, \boldsymbol{v}, \boldsymbol{e}) := \boldsymbol{e}_{(\boldsymbol{T}(\boldsymbol{v}))(d)} \ ,$$

where $\boldsymbol{e}_{(\boldsymbol{\tau})(d)}$ is the d-th component of the $\boldsymbol{\tau}$-th tuple of the stacked tuple $\boldsymbol{e}$.

Finally, define $D = g_D(Z, \boldsymbol{V})$ and $Y = g_Y(D, \boldsymbol{V}, \boldsymbol{\varepsilon}_Y)$.

By construction, we have that $(Y, D, Z, \boldsymbol{V}, \boldsymbol{\varepsilon}_Y)$ satisfies $(\mathbb{G}, \boldsymbol{\mathcal{V}}, \mathcal{F}_{\boldsymbol{V}})$ with the (g_D, g_Y) defined above, and $\operatorname{Law}((g_Y(d, \boldsymbol{V}, \boldsymbol{\varepsilon}_Y) : d \in \mathcal{D}), (g_D(z, \boldsymbol{V}) : z \in \mathcal{Z}), Z) = Q$, implying that Q satisfies $(\mathbb{G}, \boldsymbol{\mathcal{V}}, \mathcal{F}_{\boldsymbol{V}})$, as required. ∎

A.2. **Proof of Theorem 2**

First, note that the random assignment of Z implies that parent and observed data distributions are related in the following convenient way.

Lemma 5. *For any $Q \in \mathcal{M}$, P_Q depends only on Q_Z, $\{Q_{D_z} : z = \mathcal{Z}\}$, and $\{Q_{Y_d | D_z = d} : d \in \mathcal{D}, z \in \mathcal{Z}\}$.*

Proof. Let (y, d, z) be an arbitrary element of the support of (Y, D, Z). Now, we have that, for any $Q \in \mathcal{M}$,

$$\begin{aligned} P_Q\left(Y \leq y, D = d, Z = z\right) = Q\left(Y_d \leq y, D_z = d\right) Q\left(Z = z\right) \\ = Q\left(Y_d \leq y \mid D_z = d\right) Q\left(D_z = d\right) Q\left(Z = z\right) \ , \end{aligned}$$

which immediately implies the result. □

Theorem 2 (Sharp Characterization of $\mathcal{Q}_{\boldsymbol{\mathcal{T}}}$)**.** *For any $\boldsymbol{\mathcal{T}} \subseteq \mathcal{D}^{\mathcal{Z}}$, $Q \in \mathcal{Q}_{\boldsymbol{\mathcal{T}}}$ if and only if $Q \in \mathcal{M}$, $Q_Z = P_Z$ and, for each $d \in \mathcal{D}$ and $y \in \mathcal{Y}$, the following equality holds*

$$\mathbf{R}_d \boldsymbol{\pi} \boldsymbol{\xi}_{Y_d}\left(y\right) = \mathbf{pf}_d\left(y\right) \ . \tag{$\star$}$$

Proof. ($\Longleftarrow$) For any $z \in \mathcal{Z}, d \in \mathcal{D}$ and $y \in \mathcal{Y}$, we have that

$$\begin{aligned} & Q\left(Y_d \leq y, D_z = d, Z = z\right) \\ &= Q\left(Z = z\right) \int_{u \in \mathcal{Y}: u \leq y} \sum_{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}} : \boldsymbol{\tau}(z) = d} Q(\boldsymbol{T} = \boldsymbol{\tau}) f_{d,Q}(u \mid \boldsymbol{T} = \boldsymbol{\tau}) \, d\mu(u) \\ &= P\left(Z = z\right) P\left(D = d \mid Z = z\right) P\left(Y \leq y \mid D = d, Z = z\right) \\ &= P\left(Y \leq y, D = d, Z = z\right) \ , \end{aligned}$$

implying that $P_Q = P$, so $Q \in \mathcal{Q}$.

We also have that

$$\begin{aligned} & \sum_{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}} Q(\boldsymbol{T} = \boldsymbol{\tau}) \\ &= \sum_{d \in \mathcal{D}} \sum_{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}} : \boldsymbol{\tau}(z) = d} Q(\boldsymbol{T} = \boldsymbol{\tau}) \\ &= \sum_{d \in \mathcal{D}} \int_{y \in \mathcal{Y}} \sum_{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}} : \boldsymbol{\tau}(z) = d} Q(\boldsymbol{T} = \boldsymbol{\tau}) f_{d,Q}\left(y \mid \boldsymbol{T} = \boldsymbol{\tau}\right) \, d\mu(y) \\ &= \sum_{d \in \mathcal{D}} P\left(D = d \mid Z = z\right) \\ &= 1 \ , \end{aligned}$$

so $Q \in \mathcal{Q}_{\boldsymbol{\mathcal{T}}}$, as required.

($\Longrightarrow$) For any $Q \in \mathcal{M}$, we have that

$$\sum_{\boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}} : \boldsymbol{\tau}(z) = d} Q(\boldsymbol{T} = \boldsymbol{\tau})) f_{d,Q}\left(y \mid \boldsymbol{T} = \boldsymbol{\tau}\right) = P_Q\left(D = d \mid Z = z\right) f_{P_Q}\left(y \mid D = d, Z = z\right) \ .$$

Recall that $Q \in \mathcal{Q}_{\boldsymbol{\mathcal{T}}}$ implies that $P_Q = P$ and $\sum_{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}}} Q\left(D_z = \boldsymbol{\tau}(z), \, \forall z \in \mathcal{Z}\right) = 1$ for all $z \in \mathcal{Z}$. Since $\mathcal{Q}_{\boldsymbol{\mathcal{T}}} \subseteq \mathcal{M}$, we have that

$$\sum_{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}} : \boldsymbol{\tau}(z) = d} Q(\boldsymbol{T} = \boldsymbol{\tau}) f_{d,Q}\left(y \mid \boldsymbol{T} = \boldsymbol{\tau}\right) = P\left(D = d \mid Z = z\right) f_P\left(y \mid D = d, Z = z\right) \ ,$$

which immediately implies the result. □

A.3. Proof of Theorem 3

Theorem 3. *For any* $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$,

$$\Pi_{\mathcal{T}} = \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d} \,,$$

and $\Pi_{\mathcal{T}}$ *is a compact convex polytope in* $\mathbb{R}^{|\mathcal{T}|}_{\geq 0}$. *Moreover,* $\mathcal{Q}_{\mathcal{T}} \neq \emptyset$ *if and only if* $\Pi_{\mathcal{T}} \neq \emptyset$.

Proof. First, assume that $\mathcal{Q}_{\mathcal{T}} \neq \emptyset$, which implies that $\Pi_{\mathcal{T}} \neq \emptyset$, and fix a vector $p \in \Pi_{\mathcal{T}}$. Now, we must have that there exists $Q \in \mathcal{Q}_{\mathcal{T}}$ such that $\pi_Q = p$, and

$$\sum_{\boldsymbol{\tau} \in \mathcal{T} : \boldsymbol{\tau}(z) = d} \pi_Q(\boldsymbol{\tau}) f_{d,Q}\left(y \mid T = \boldsymbol{\tau}\right) = p_{(d|z)} f_{(d,z)}(y) \quad \forall (d, y, z) \in \mathcal{D} \times \mathcal{Y} \times \mathcal{Z} \,.$$

For any $d \in \mathcal{D}$, define $g^{(d)}_{(\boldsymbol{\tau})}(y) := \pi_Q(\boldsymbol{\tau}) f_{d,Q}\left(y \mid T = \boldsymbol{\tau}\right)$ for each $y \in \mathcal{Y}$, and note that $g^{(d)} \in \Gamma\left(\mathcal{B}_d\right)$ and $\int_{\mathcal{Y}} g^{(d)}(y)\ d\mu(y) = \pi_Q = p$, and so, $p \in \Pi_{\mathcal{T} \upharpoonright d}$. This works for any d, and so, $\Pi_{\mathcal{T}} \subseteq \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T} \upharpoonright d}$. Note that we now also have that

$$\mathcal{Q}_{\mathcal{T}} \neq \emptyset \implies \Pi_{\mathcal{T}} \neq \emptyset \implies \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T} \upharpoonright d} \neq \emptyset \,.$$

Next, assume that $\bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T} \upharpoonright d} \neq \emptyset$, fix a $p \in \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T} \upharpoonright d}$, and note that $p \in \varpi_{\mathcal{T}}$ trivially. Now, we have that, for each $d \in \mathcal{D}$, there exists $g^{(d)} \in \Gamma\left(\mathcal{B}_d\right)$ such that $\int_{\mathcal{Y}} g^{(d)}(y)\ d\mu(y) = p$. Define, for each $d \in \mathcal{D}$ and $\boldsymbol{\tau} \in \mathcal{T}$,

$$f^{(d)}_{\boldsymbol{\tau}}(y) := \frac{g^{(d)}_{(\boldsymbol{\tau})}(y)}{p_{(\boldsymbol{\tau})}} = \frac{g^{(d)}_{(\boldsymbol{\tau})}(y)}{\int_{\mathcal{Y}} g^{(d)}_{(\boldsymbol{\tau})}(y)\ d\mu(y)} .$$

Now, $f^{(d)}_{\boldsymbol{\tau}}(y)$ is a valid density function for each $\boldsymbol{\tau} \in \mathcal{T}$ and each $d \in \mathcal{D}$; let $F^{(d)}_{\boldsymbol{\tau}}(y)$ be the CDF associated with this density.

For each $\boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}}$, let $C_{\boldsymbol{\tau}}$ be some $|\mathcal{D}|$-dimensional copula and, when $\boldsymbol{\tau} \notin \mathcal{T}$, define $p_{(\boldsymbol{\tau})} := 0$. Now, define Q as

$$Q\left(y, \boldsymbol{\tau}, z\right) := C_{\boldsymbol{\tau}}\left(\left(F^{(d)}_{\boldsymbol{\tau}}\left(y_{(d)}\right) : d \in \mathcal{D}\right)\right) \times p_{(\boldsymbol{\tau})} \times Q_Z\left(z\right) \;,$$

for any $y \in \prod_{d \in \mathcal{D}} \mathcal{Y}$, $\boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}}$, and $z \in \mathcal{Z}$.

This construction ensures that $Q \in \mathcal{M}$ and, by Lemma 5, that $P_Q = P$. Since $\pi_Q = p$ by construction, and p only has support in $\mathcal{T}$, we have that $Q \in \mathcal{Q}_{\mathcal{T}}$, which implies that $\Pi_{\mathcal{T}} \supseteq \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T} \upharpoonright d}$ and $\mathcal{Q}_{\mathcal{T}} \neq \emptyset$, completing the proof. □

A.4. Proofs of Lemmas 1 and 2

A.4.1. *Proof of Lemma 1.*

Proof. First consider the map $y \mapsto \tilde{\mathcal{B}}_d\left(y\right)$ where $\tilde{\mathcal{B}}_d\left(y\right) := \left\{ \boldsymbol{x} \in \mathbb{R}^{|\mathcal{T}_d|} \;\middle|\; \tilde{\mathbf{R}}_d \boldsymbol{x} = \mathbf{pf}_d\left(y\right) \right\}$, where $\mathcal{T}_d := \mathcal{T} \setminus \left\{\boldsymbol{\tau} \in \mathcal{T} \mid \boldsymbol{\tau}\left(z\right) \neq d, \forall z \in \mathcal{Z}\right\}$ and $\tilde{\mathbf{R}}_d$ is $\mathbf{R}_d$ with its zero columns removed. Now, $\tilde{\mathcal{B}}_d\left(y\right)$ is a compact convex polytope, and so, $y \mapsto \tilde{\mathcal{B}}_d\left(y\right)$ is a polytope bundle, as defined in Billera and Sturmfels (1992). Let $\int \tilde{\mathcal{B}}_d$ be the Minkowski integral of this bundle, and note that Proposition 1.2 in Billera and Sturmfels (1992) implies that, for any $\boldsymbol{c} \in \mathbb{R}^{|\mathcal{T}_d|}$,

$h_{\int \tilde{\mathcal{B}}_d}(\boldsymbol{c}) = \int_{\mathcal{Y}} h_{\tilde{\mathbf{R}}_d}(\boldsymbol{c};\ \mathbf{pf}_d(y))\ d\mu(y)$. Note that if $\boldsymbol{\tau} \in \{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}} \mid \boldsymbol{\tau}(z) \neq d, \forall z \in \mathcal{Z}\}$, then for any $\boldsymbol{c} \in \mathbb{R}^{|\boldsymbol{\mathcal{T}}|}$, $\boldsymbol{c}_{(\boldsymbol{\tau})} > 0$ implies that

$$h_{\Pi_{\boldsymbol{\mathcal{T}}|d}}(\boldsymbol{c}) = \infty = \int_{\mathcal{Y}} \infty\ d\mu(y) = \int_{\mathcal{Y}} h_{\mathbf{R}_d}(\boldsymbol{c};\ \mathbf{pf}_d(y))\ d\mu(y)\ ,$$

and if $\boldsymbol{c}_{(\boldsymbol{\tau})} \leq 0$ for all $\boldsymbol{\tau} \in \{\boldsymbol{\tau} \in \boldsymbol{\mathcal{T}} \mid \boldsymbol{\tau}(z) \neq d, \forall z \in \mathcal{Z}\}$, then letting $\tilde{\boldsymbol{c}}$ be the projection of $\boldsymbol{c}$ onto $\mathbb{R}^{|\boldsymbol{\mathcal{T}}_d|}$, we have

$$h_{\Pi_{\boldsymbol{\mathcal{T}}|d}}(\boldsymbol{c}) = h_{\int \tilde{\mathcal{B}}_d}(\tilde{\boldsymbol{c}}) = \int_{\mathcal{Y}} h_{\tilde{\mathbf{R}}_d}(\tilde{\boldsymbol{c}};\ \mathbf{pf}_d(y))\ d\mu(y) = \int_{\mathcal{Y}} h_{\mathbf{R}_d}(\boldsymbol{c};\ \mathbf{pf}_d(y))\ d\mu(y)\ ,$$

which immediately implies the first claim.

For the second claim: Since $\mathbf{R}_d$ does not depend on y, Theorem 1.3 in Billera and Sturmfels (1992) (under the "finite measurable subdivision" definition of piecewise-linear bundle) implies that $\int \tilde{\mathcal{B}}_d$ is a compact convex polytope in $\mathbb{R}_{\geq 0}^{|\boldsymbol{\mathcal{T}}_d|}$, which immediately implies that $\Pi_{\boldsymbol{\mathcal{T}}|d}$ is a convex polytope, since $\Pi_{\boldsymbol{\mathcal{T}}|d}$ is simply the representation of $\int \tilde{\mathcal{B}}_d$ in $\mathbb{R}_{\geq 0}^{|\boldsymbol{\mathcal{T}}|}$. □

A.4.2. *Proof of Lemma 2.*

Proof. Define $\mathcal{E}_d(y) := \left\{ (\boldsymbol{x}_1, \boldsymbol{x}_2) \in \mathbb{R}^{|\boldsymbol{\mathcal{T}}|} \times \mathbb{R}_{\geq 0}^{|\boldsymbol{\mathcal{T}}|} \,\middle|\, \boldsymbol{x}_1 = \kappa(y)\,\boldsymbol{x}_2,\ \mathbf{R}_d \boldsymbol{x}_2 = \mathbf{pf}_d(y) \right\}$
By the same argument as Lemma 1, we have that

$$h_{\Theta_{\boldsymbol{\mathcal{T}}|d}(d)}(\mathbf{c}_1, \mathbf{c}_2) = \int_{\mathcal{Y}} h_{\mathcal{E}_d(y)}(\mathbf{c}_1, \mathbf{c}_2)\ d\mu(y)\ .$$

The result now follows by noting that

$$\begin{aligned} h_{\mathcal{E}_d(y)}(\boldsymbol{c}_1, \boldsymbol{c}_2) :=& \sup_{(\boldsymbol{x}_1, \boldsymbol{x}_2) \in \mathcal{E}_d(y)} \boldsymbol{c}_1^{\intercal} \boldsymbol{x}_1 + \boldsymbol{c}_2^{\intercal} \boldsymbol{x}_2 \\ =& \sup_{(\boldsymbol{x}_1, \boldsymbol{x}_2) \in \mathcal{E}_d(y)} \boldsymbol{c}_1^{\intercal} \kappa(y)\, \boldsymbol{x}_2 + \boldsymbol{c}_2^{\intercal} \boldsymbol{x}_2 \\ =& \sup_{\boldsymbol{x} \in \mathcal{B}_d(\mathbf{pf}_d(y))} (\kappa(y)\, \boldsymbol{c}_1 + \boldsymbol{c}_2)^{\intercal} \boldsymbol{x} \\ =& h_{\mathbf{R}_d}(\kappa(y)\, \boldsymbol{c}_1 + \boldsymbol{c}_2;\ \mathbf{pf}_d(y))\ . \end{aligned}$$ □

APPENDIX B: PROOFS OF INFERENCE RESULTS

B.1. Proofs of Results related to Specification Test

Proof of Lemma 3.

Proof. Here, $\mathbf{R}_d = \mathbf{R}_{d, \boldsymbol{\mathcal{T}}(\mathbb{S})}$.
We have that $\mathcal{B}_{\mathbf{R}_d}(\mathbf{pf}_d) \neq \emptyset \iff \mathbf{pf}_d(y) \in \mathrm{cone}(\mathbf{R}_d)$.
By the Minkowski-Weyl theorem, every finite cone is a polyhedral cone, i.e. there exists a matrix $\mathbf{A}_d$ such that

$$\{\boldsymbol{x} : \exists \boldsymbol{x} \geq 0, \mathbf{R}_d \boldsymbol{x} = b\} = \{\boldsymbol{x} : \mathbf{A}_d b \leq 0\}\ .$$

The complexity of finding such a matrix $\mathbf{A}_d$ is the same as converting the vertex-representation of a convex polytope to its halfspace representation (indeed, applying such an algorithm to the set of columns of $\mathbf{R}_d$ will yield $\mathbf{A}_d$).

Now, we have that $\mathcal{B}_{\mathbf{R}_d}(\mathbf{pf}_d) \neq \emptyset \iff \mathbf{A}_d \mathbf{pf}_d(y) \leq 0$. This restriction represents a set of linear inequalities, so that, for $j \in [|\operatorname{rows}(\mathbf{A}_d)|]$, the linear inequality represented by the j-th row is given by

$$\begin{aligned}
&\sum_{z\in\mathcal{Z}} \tilde{c}_{z,j} P(D=d \mid Z=z) f_{P;Y}(y \mid D=d, Z=z) \\
&= \sum_{z\in\mathcal{Z}} \tilde{c}_{z,j} \frac{P(D=d, Z=z)}{P(Z=z)} f_{P;Y}(y \mid D=d, Z=z) \leq 0 \,,
\end{aligned}$$

for some coefficients $\{\tilde{c}_{z,j} : z \in \mathcal{Z}\}$.

Define $c_{z,j} := \tilde{c}_{z,j}\left(\prod_{z'\in\mathcal{Z}:z'\neq z} P\left(Z=z'\right)\right)$. Next, note that

$$\begin{aligned}
&\sum_{z\in\mathcal{Z}} \tilde{c}_{z,j} \frac{P(D=d, Z=z)}{P(Z=z)} f_{P;Y}(y \mid D=d, Z=z) \leq 0, \text{ for all } y \in \mathcal{Y} \\
\iff &\frac{1}{\prod_{z\in\mathcal{Z}} P(Z=z)} \sum_{z\in\mathcal{Z}} c_{z,j} P(D=d, Z=z) f_{P;Y}(y \mid D=d, Z=z) \leq 0, \text{ for all } y \in \mathcal{Y} \\
\iff &\sum_{z\in\mathcal{Z}} c_{z,j} P(D=d, Z=z) f_{P;Y}(y \mid D=d, Z=z) \leq 0, \text{ for all } y \in \mathcal{Y} \\
\iff &\sum_{z\in\mathcal{Z}} c_{z,j} P(D=d, Z=z \mid Y=y) f_{P;Y}(y) \leq 0, \text{ for all } y \in \mathcal{Y} \\
\iff &\sum_{z\in\mathcal{Z}} c_{z,j} P(D=d, Z=z \mid Y=y) \leq 0, \text{ for all } y \in \mathcal{Y} \\
\iff &\sum_{z\in\mathcal{Z}} c_{z,j} E\left[\mathbb{1}\{D=d, Z=z\} \mid Y=y\right] \leq 0, \text{ for all } y \in \mathcal{Y} \\
\iff &E\left[\sum_{z\in\mathcal{Z}} c_{z,j} \mathbb{1}\{D=d, Z=z\} \,\middle|\, Y=y\right] \leq 0, \text{ for all } y \in \mathcal{Y}
\end{aligned}$$

The result now follows by applying the same argument to each row of $\mathbf{A}_d$, and to each $d \in \mathcal{D}$. □

Proof of Lemma 4.

Proof. First assume that $\mathcal{B}_d(\mathbf{pf}_d(y)) \neq \emptyset$, for all $y \in \mathcal{Y}$ and $d \in \mathcal{D}$.

Define $\boldsymbol{\mathcal{T}}_d := \{\tau \in \boldsymbol{\mathcal{T}} : \boldsymbol{\tau}(z) = d \text{ for some } z \in \mathcal{Z}\}$. (i.e. all response-types in $\boldsymbol{\mathcal{T}}$ except d-Never-Takers). Assume that $|\boldsymbol{\mathcal{T}}_d| > |\mathcal{Z}|$ (the case $|\boldsymbol{\mathcal{T}}_d| < |\mathcal{Z}|$ is simpler), and that $\mathbf{R}_d$ has full row-rank (to simplify exposition). Assume WLOG that $\mathbf{R}_d$ is such that the d-Never-Takers are the last $|\boldsymbol{\mathcal{T}} \setminus \boldsymbol{\mathcal{T}}_d|$ columns of $\mathbf{R}_d$.

The polytope $\mathcal{B}_d(\mathbf{pf}_d)$ (which is in $\mathbb{R}^{|\boldsymbol{\mathcal{T}}|}$) as a polytope in $\mathbb{R}^{|\boldsymbol{\mathcal{T}}_d|-|\mathcal{Z}|}$ defined by all solutions to the system of inequalities

$$\mathbf{M}_d \tilde{\boldsymbol{x}} \leq \boldsymbol{b}_d(y) \ ,$$

for some matrix $\mathbf{M}_d$ and vector $\boldsymbol{b}_d(y)$ which can be obtained via row-reduction and elimination of pivot variables.

Denote by $\mathbf{M}_{d,(i)}$ the i-th row of $\mathbf{M}_d$, and define for each row, i, of $\mathbf{M}_d$,

$$h_{\mathcal{B}_d}\left(\mathbf{M}_{d,(i)}, \boldsymbol{b}_d(y)\right) := \sup_{\boldsymbol{x}:\mathbf{M}_d\boldsymbol{x}\leq\boldsymbol{b}_d(y)} \left\{\mathbf{M}_{d,(i)}^{\intercal}\boldsymbol{x}\right\} = h_{\mathbf{R}_d}\left(\mathbf{M}_{d,(i)};\ \mathbf{pf}_d\left(y\right)\right) \ ,$$

where in the last equality we represent $\mathbf{M}_{d,(i)}$ as a vector in $\mathbb{R}^{\mathcal{T}}$ (by setting extra coordinates to 0). Now note that the solutions to

$$\mathbf{M}_d\tilde{\boldsymbol{x}} \leq \begin{bmatrix} h_{\mathcal{B}_d}\left(\mathbf{M}_{d,(1)}, \boldsymbol{b}_d(y)\right) \\ \vdots \\ h_{\mathcal{B}_d}\left(\mathbf{M}_{d,(n_d)}, \boldsymbol{b}_d(y)\right) \end{bmatrix} \ ,$$

where n_d is the number of rows in $\mathbf{M}_d$, represents the same polytope as the solutions to $\mathbf{M}_d\tilde{\boldsymbol{x}} \leq \boldsymbol{b}_d\left(y\right)$. The result now follows by repeating this argument for each $d \in \mathcal{D}$, applying Theorem 1.2 in Billera and Sturmfels (1992) along with Proposition 10 to obtain the halfspace representation of $\Pi_{\boldsymbol{\mathcal{T}}|d}$, and then applying Theorem 3 with the halfspace representation of $\mathcal{F}_{\boldsymbol{T}}\left(\mathbb{S}\right)$ given by $\mathbf{S}\boldsymbol{x} \leq \mathbf{s}$, where such a matrix $\mathbf{S}$ and vector $\mathbf{s}$ must exist since $\mathbb{S}$ satisfies Condition **C**.

Finally, the case where $\mathcal{B}_d\left(\mathbf{pf}_d\left(y\right)\right) = \emptyset$ for some $y \in \mathcal{Y}$ and $d \in \mathcal{D}$ follows automatically for the definitions of $\mathbf{M}_d$, $\mathbf{S}$, and $\boldsymbol{s}$ given above, since $h_{\mathbf{R}_d}\left(\mathbf{M}_{d,(i)};\ \mathbf{pf}_d\left(y\right)\right) = -\infty$ for all y in some subset with positive measure implies $\int_{\mathcal{Y}} h_{\mathbf{R}_d}\left(\mathbf{M}_{d,(i)};\ \mathbf{pf}_d\left(y\right)\right)\ d\mu\left(y\right) = -\infty$. □

Proof of Proposition 7.

Proof. Assume that $\mathcal{B}_d\left(\mathbf{pf}_d\left(y\right)\right) \neq \emptyset$, for all $y \in \mathcal{Y}$ and $d \in \mathcal{D}$ (if not, the result follows immediately from Lemma 3), so that $\boldsymbol{h}_{\mathbb{S}}$ is a real vector. Let the system of inequalities given by Lemma 4 be denoted $\mathbf{M}\boldsymbol{x} \leq \boldsymbol{h}_{\mathbb{S}}$. This system has a solution iff $\boldsymbol{h}_{\mathbb{S}} \in \mathrm{cone}\left(\mathbf{M}\right)$, and so, by the Minkowski-Weyl theorem, there exists a matrix $\mathbf{A}$ such that the system has a solution iff $\mathbf{A}\boldsymbol{h}_{\mathbb{S}} \leq 0$. As mentioned in the proof of Lemma 3, the complexity of finding $\mathbf{A}$ is the same as converting a convex polytope from its vertex representation to its halfspace representation. The result now follows immediately from Lemmas 3 and 4 by defining $\boldsymbol{c}_i := \mathbf{A}_i$. □

B.2. **Proof of Proposition 8**

Proof. First, note that

$$\Theta_{\mathbb{S}}\left(\beta\right) = \left\{b \in \mathbb{R}\ \middle|\ \Lambda_{\mathbb{S}}\left(\beta; b\right) \cap \left(\Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}\left(\kappa\right) \times \Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d'}\left(\kappa\right)\right) \neq \emptyset\right\} \ .$$

Next, define $\Theta := \Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}\left(\kappa\right) \times \Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d'}\left(\kappa\right)$ and note that the support function of Θ is just the sum of the support functions of $\Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d}\left(\kappa\right)$ and $\Theta_{\boldsymbol{\mathcal{T}}(\mathbb{S})|d'}\left(\kappa\right)$, evaluated at their respective arguments:

$$\begin{aligned} h_{Theta}\left(\boldsymbol{c}_{(d,1)}, \boldsymbol{c}_{(d,2)}, \boldsymbol{c}_{(d',1)}, \boldsymbol{c}_{(d',2)}\right) &:= h_{\Theta_{\boldsymbol{\mathcal{T}}|d}(\kappa)}\left(\boldsymbol{c}_{(d,1)}, \boldsymbol{c}_{(d,2)}\right) + h_{\Theta_{\boldsymbol{\mathcal{T}}|d'}(\kappa)}\left(\boldsymbol{c}_{(d',1)}, \boldsymbol{c}_{(d',2)}\right) \\ &= \int_{\mathcal{Y}} h_{\mathbf{R}_d}\left(\kappa\left(y\right)\boldsymbol{c}_{(d,1)} + \boldsymbol{c}_{(d,2)}\ ;\ \mathbf{pf}_d\left(y\right)\right)\ d\mu\left(y\right) \\ &\quad + \int_{\mathcal{Y}} h_{\mathbf{R}_{d'}}\left(\kappa\left(y\right)\boldsymbol{c}_{(d',1)} + \boldsymbol{c}_{(d',2)}\ ;\ \mathbf{pf}_{d'}\left(y\right)\right)\ d\mu\left(y\right) \ . \end{aligned} \tag{8}$$

Note that Θ is always a non-empty closed convex set when $\mathcal{Q}_{\mathbb{S}} \neq \emptyset$. Now, WLOG[20]

$$\gamma \notin \Theta_{\mathbb{S}}\left(\beta\right)$$

[20] If γ is such that $\Lambda_{\mathbb{S}}\left(\beta; \gamma\right)$ is non-empty, the test automatically rejects the null (as desired) since the last term will resolve to $-\left(-\infty\right) = \infty$, making the whole expression resolve to ∞ for any $\boldsymbol{v}$.

$$
\begin{aligned}
&\iff \Lambda_{\mathbb{S}}(\beta;\gamma)\cap\Theta=\emptyset \\
&\iff \exists\, \boldsymbol{v}\in S^{4|\mathbf{T}|-1},\ \sup_{\boldsymbol{x}\in\Theta}\boldsymbol{x}^{\intercal}\boldsymbol{v} < \inf_{\boldsymbol{x}\in\Lambda_{\mathbb{S}}(\beta;\gamma)}\boldsymbol{x}^{\intercal}\boldsymbol{v} \\
&\iff \exists\, \boldsymbol{v}\in S^{4|\mathbf{T}|-1},\ h_{\Theta}(\boldsymbol{v}) < h_{\Lambda_{\mathbb{S}}(\beta;\gamma)}(-\boldsymbol{v}) \\
&\iff \exists\, \boldsymbol{v}\in S^{4|\mathbf{T}|-1},\ 0 < -h_{\Theta}(\boldsymbol{v}) - h_{\Lambda_{\mathbb{S}}(\beta;\gamma)}(-\boldsymbol{v}) \\
&\iff \sup_{\boldsymbol{v}\in S^{4|\mathbf{T}|-1}}\left(-h_{\Theta}(\boldsymbol{v}) - h_{\Lambda_{\mathbb{S}}(\beta;\gamma)}(-\boldsymbol{v})\right) > 0\,,
\end{aligned}
$$

and the result immediately follows by plugging in (8) for h_{Θ}. □

APPENDIX C: COMPUTATIONAL DETAILS

C.1. Proof of Proposition 9

Proof. (Sorry, this proof is unnecessarily complicated - will fix in future drafts.)

WLOG take $\delta=0$ since it implies the result for all $\delta>0$. By $\mathcal{B}_d(\mathbf{pf}_d(y))\neq\emptyset$ for all $y\in\mathcal{Y}$, we have that $\mathbf{pf}_d(y)\in\operatorname{cone}(\mathbf{R}_d)$ for all y - therefore, the first part of the generic Step j (finding a feasible basis) will terminate. In the remainder, I focus on the second part of the generic Step j (finding an optimal basis).

First, partition $\mathcal{Y}$ into $\mathcal{Y}_1,\ldots,\mathcal{Y}_N$ such that each $y,y'\in\mathcal{Y}_i \implies \mathcal{B}_d(\mathbf{pf}_d(y))$ is normally equivalent to $\mathcal{B}_d(\mathbf{pf}_d(y'))$. Since $\mathbf{pf}_d$ is continuous (and $\mathcal{Y}$ is compact), we can take the partition to be defined by numbers

$$\min\mathcal{Y} =: y_0 \leq \cdots \leq y_N := \max\mathcal{Y}\,,$$

such that[21]

$$\mathcal{Y} = [y_0,y_1]\sqcup\left(\bigsqcup_{i=1}^{N-1}(y_i,y_{i+1}]\right)\,.$$

By definition, for y within each partition $\mathcal{Y}_k$, $\mathcal{B}_d(\mathbf{pf}_d(y))$ has the same normal fan, denote it $\mathcal{N}_k$, which is a decomposition of $\mathbb{R}^{|\boldsymbol{\mathcal{T}}|}$ into polyhedral cones, $\mathcal{N}_{k,1},\ldots,\mathcal{N}_{k,N_k}$.

Now, further partition each cell $\mathcal{Y}_k$ of the above partition into $\mathcal{Y}_{k,1},\ldots,\mathcal{Y}_{k,N_k}$ such that $y,y'\in\mathcal{Y}_{k,j}$ implies that $\varkappa(y),\varkappa(y')\in\mathcal{N}_{k,j}$. Since $\varkappa$ is a continuous function over a compact set, we can once again WLOG take these partitions to be intervals [22].

Now, we have partitioned $\mathcal{Y}$ into finitely many intervals such that the LP

$$\max\ \varkappa(y)^{\intercal}\boldsymbol{x} \quad \text{subject to} \quad \mathbf{R}_d\boldsymbol{x}=\mathbf{pf}_d(y)\,,\ \boldsymbol{x}\geq\mathbf{0}\,,$$

will have the same optimal base(s) for all y within each cell of the partition. This shows the final claim.

Now, it suffices to show that there will always exist some $n<\infty$ such that, for all cells (k,j), either $\mathcal{G}_n\cap\operatorname{int}(\mathcal{Y}_{k,j})\neq\emptyset$ or $\mathcal{G}_n\cap\mathcal{Y}_{k,j}\neq\emptyset$ and the LP has the same optimal base for $y\in\mathcal{G}_n\cap\mathcal{Y}_{k,j}$ as it does for points on the interior.

This follows immediately by noting that if n is such that this condition does not hold for some cell(s), then points in all such cells will have positive duality gap, and so, $\mathcal{G}_{n+1}$ will

[21]Technical point: to represent the partitions in this way, we can relax the requirement of normal equivalence for points on the boundary, so that they are only required to coarsen the normal fan associated with the interior points of the partition. The intuition is the same, however.

[22]This may lead to potentially more than N_k partitions for each $\mathcal{Y}_k$, but simply redefine N_k to accommodate this, if necessary.

necessary satisfy the condition for at least one more cell. Since there are only finitely many cells, this argument implies termination in finitely many steps. This shows the first claim in (i). (ii) and the second claim in (i) are immediate consequences of the weak duality theorem, completing the proof. $\square$

APPENDIX D: PROOFS OF AUXILIARY RESULTS

Proposition 10. *Let P_1 and P_2 be bounded full-dimensional polytopes in $\mathbb{R}^n$ defined as $\{x \geq 0 : Ax \leq b_1\}$ and $\{x \geq 0 : Ax \leq b_2\}$, respectively. Denote by h_{P_i} the support function of P_i. Let $\{a_1, \ldots, a_N\}$ be the rows of A, let $\bar{b}_1 := (h_{P_1}(a_i) : i \in \{1, \ldots, N\})$, and let $\bar{b}_2 := (h_{P_2}(a_i) : i \in \{1, \ldots, N\})$. Then,*

$$P_1 + P_2 = \left\{x \geq 0 : Ax \leq \bar{b}_1 + \bar{b}_2\right\} .$$

Proof. Denote $\bar{P} := \left\{x \geq 0 : Ax \leq \bar{b}_1 + \bar{b}_2\right\}$.
First, note that $h_{P_1+P_2}(a) = h_{P_1}(a) + h_{P_2}(a)$ for all $a \in \mathbb{R}^n$, and so, $P_1 + P_2 \subseteq \bar{P}$.
Next, let ∂P denote the boundary of P, for any polytope P. We have that

$$\partial (P_1 + P_2) = \bigcup_{i=1}^{N} \left\{x_1 + x_2 : x_1 \in P_1, x_2 \in P_2; a_i' x_1 = h_{P_1}(a_i), a_i' x_2 = h_{P_2}(a_i)\right\} .$$

Now, note that if $(x_1 + x_2) \in \partial (P_1 + P_2)$ such that $a_i' x_1 = h_{P_1}(a_i), a_i' x_2 = h_{P_2}(a_i)$, then necessarily $a_i' (x_1 + x_2) = h_{P_1}(a_i) + h_{P_2}(a_i) = \bar{b}_1 + \bar{b}_2$, and so, $x_1 + x_2 \in \partial \bar{P}$. Therefore, we have that $\partial (P_1 + P_2) \subseteq \partial \bar{P}$.

Next, note that both $P_1 + P_2$ and $\bar{P}$ have the same dimension, n, and so, their boundaries are both homeomorphic to S^{n-1}, the unit sphere in $\mathbb{R}^{n-1}$[23]. Let f be the homeomorphism between $\bar{P}$ and S^{n-1}.

Suppose that $\partial (P_1 + P_2) \neq \partial \bar{P}$, then $f_{\restriction \partial (P_1+P_2)}$ is a homeomorphism between $\partial (P_1 + P_2)$ and $f(\partial (P_1 + P_2)) \subsetneq S^{n-1}$. Since $\partial (P_1 + P_2)$ is itself homeomorphic to S^{n-1}, this implies that S^{n-1} is homeomorphic to its strict subset $f(\partial (P_1 + P_2))$. This is a contradiction, however, since the only subset of S^{n-1} that is homeomorphic to S^{n-1} is S^{n-1} itself. Therefore, $\partial (P_1 + P_2) = \partial \bar{P}$, which immediately implies that $\bar{P} = P_1 + P_2$, as required[24]. $\square$

Corollary 3. *Let P_1 and P_2 be bounded full-dimensional polytopes in $\mathbb{R}^n$ defined as $\{x \geq 0 : Ax \leq b_1\}$ and $\{x \geq 0 : Bx \leq b_2\}$, respectively. Let $A = \begin{pmatrix} A' \\ C \end{pmatrix}$ and let $B = \begin{pmatrix} B' \\ C \end{pmatrix}$, where C is the matrix consisting of (potentially zero) rows that are shared between A and B. Let $\{d_1, \ldots, d_N\}$ be the rows of $\begin{pmatrix} A' \\ B' \\ C \end{pmatrix} =: D$, and let $\bar{b}_1 := (h_{P_1}(d_i) : i \in \{1, \ldots, N\})$, and let $\bar{b}_2 := (h_{P_2}(d_i) : i \in \{1, \ldots, N\})$. Then,*

$$P_1 + P_2 = \left\{x \geq 0 : Dx \leq \bar{b}_1 + \bar{b}_2\right\} .$$

Proof. Follows immediately from Proposition 10 by noting that

$$P_1 = \left\{x \geq 0 : Dx \leq \bar{b}_1\right\} \text{ and } P_2 = \left\{x \geq 0 : Dx \leq \bar{b}_2\right\} . \square$$

[23] If $K \subseteq \mathbb{R}^n$ is convex, compact, and has non-empty interior, then its boundary is homeomorphic to S^{n-1}.
[24] Special thanks to Eric Wofsey for the idea used to prove $\partial (P_1 + P_2) \subseteq \partial \bar{P} \implies \partial (P_1 + P_2) = \partial \bar{P}$.

APPENDIX E: EQUIVALENCE OF GENERALIZED ROY MODEL AND NEYMAN-RUBIN POTENTIAL OUTCOMES MODEL

I assume that the probability space $(\Omega, \Sigma, \mathrm{P})$ is sufficiently enriched for the applications of the Skorohod's theorem used in this section. This assumption can always be satisfied after suitably enriching $(\Omega, \Sigma, \mathrm{P})$, if needed, by taking product spaces[25].

Condition PO (Neyman-Rubin Potential Outcomes Model)**.** *There exists random variables* $((Y_{d,z} : d \in \mathcal{D}, z \in \mathcal{Z}), (D_z : z \in \mathcal{Z}))$ *defined on* $(\Omega, \Sigma, \mathrm{P})$ *such that:*

(i) *For all* $z, z' \in \mathcal{Z}$ *and* $d \in \mathcal{D}$, $Y_d := Y_{d,z} = Y_{d,z'}$ *almost surely. Further,*

$$D = \sum_{z \in \mathcal{Z}} \mathbb{1}\{Z = z\} D_z ,$$

$$Y = \sum_{d \in \mathcal{D}} \mathbb{1}\{D = d\} Y_d ,$$

almost surely.

(ii) $((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z})) \perp Z$

Proposition 11. *Let* P *be a distribution of the observed data,* (Y, D, Z). P *satisfies Condition* ***PO*** *if and only if it satisfies Condition* ***M****.*

Proof. First, suppose P satisfies Condition **M**. Define

$$Y_d := f_Y(d, \mathbf{V}, \varepsilon_Y) ,$$

$$D_z := f_D(z, \mathbf{V}) ,$$

and note that Condition **M**(ii) implies that

$$\begin{aligned} & Z, \mathbf{V}, \varepsilon_Y \text{ are mutually independent} \\ \implies & ((f_Y(d, \mathbf{V}, \varepsilon_Y) : d \in \mathcal{D}), (f_D(z, \mathbf{V}) : z \in \mathcal{Z})) \perp Z \\ \implies & ((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z})) \perp Z , \end{aligned}$$

which implies that P satisfies Condition **PO**.

Now, suppose P satisfies Condition **PO**, and let $((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z})) \sim Q$ be random variables that witness this; note that $P_Q = P$. Define $\boldsymbol{V} = (D_z : z \in \mathcal{Z})$. For each $\boldsymbol{v} \in \mathrm{supp}_Q(\boldsymbol{V})$, define $\boldsymbol{\varepsilon}_Y(\boldsymbol{v})$ as a $|\mathcal{D}|$-dimensional random vector with distribution $Q_{(Y_d : d \in \mathcal{D}) | \boldsymbol{V} = \boldsymbol{v}}$, with $\boldsymbol{\varepsilon}_Y := \left(\boldsymbol{\varepsilon}_Y(\boldsymbol{v})^\intercal : \boldsymbol{v} \in \mathrm{supp}_Q(\boldsymbol{V})\right)^\intercal$ independent from $(Z, \boldsymbol{V})$. As a corollary of the Skorohod representation theorem, we can take $\boldsymbol{\varepsilon}_Y$ to be such that, for every $\boldsymbol{v} \in \mathrm{supp}(\boldsymbol{V})$, $\mathbb{1}\{\boldsymbol{V} = \boldsymbol{v}\} \boldsymbol{\varepsilon}_Y(\boldsymbol{v}) = (\mathbb{1}\{\boldsymbol{V} = \boldsymbol{v}\} Y_d : d \in \mathcal{D})$ almost surely.

Next, define $g_D : \mathcal{Z} \times \mathrm{supp}_Q(\boldsymbol{V})$ and $g_Y : \mathcal{D} \times \mathrm{supp}(\boldsymbol{V}) \times \mathrm{supp}(\boldsymbol{\varepsilon}_Y)$ as follows

$$g_D(z, \boldsymbol{v}) := \boldsymbol{v}_{(z)} ,$$

$$g_Y(d, \boldsymbol{v}, \boldsymbol{e}) := \boldsymbol{e}_{(\boldsymbol{v})(d)} ,$$

where $\boldsymbol{e}_{(\boldsymbol{v})(d)}$ is the d-th component of the $\boldsymbol{v}$-th tuple of the stacked tuple $\boldsymbol{e} \in \mathrm{supp}(\boldsymbol{\varepsilon}_Y)$. Now, the result follows immediately by noting that $(g_Y(g_D(Z, \boldsymbol{V}), \boldsymbol{V}, \boldsymbol{\varepsilon}_Y), g_D(Z, \boldsymbol{V}), Z) = (Y, D, Z)$ almost surely. □

[25] see Halmos (1976), p. 157